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Mathematics K–10 (2022) Syllabus Support

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Generated Oct 2025

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Mathematics K–10 (2022)

Teaching advice for Stage 4

Teaching advice for Computation with integers

Content

- Computation with integers

Teaching advice

Key ideas

Students:

- locate positive and negative integers on a number line
- order integers
- compare integers using inequality symbols
- apply the 4 operations to positive and negative integers
- solve problems by calculating with integers and order of operations

Key terms

- division
- magnitude
- integers
- addition
- subtraction
- multiplication
- operation

Background knowledge

- Repeated addition
- Order of operations
- Number lines
- Number patterns

Making connections

Connections	Outcome or Stage
Operations involving positive and negative integers are used in algebraic expressions, formulae and equations.	Stage 4 Algebraic Techniques
Integers are used to identify and plot points on the Cartesian plane.	Stage 4 Linear Relationships
Operations involving positive and negative integers are used in finance, time, temperature and motion.	Stage 6 Num CEC 1.1 Stage 6 Num CEC 3.3.3 Stage 6 Num CEC 3.4.3
Addition and subtraction of positive and negative integers can be applied to UTC and time differences.	Stage 6 Num CEC 1.4 Stage 6 Num CEC 3.4.2 Stage 6 Mathematics Standard 2
Integers are used in measuring the temperature of substances and balancing chemical equations.	Stage 4 Science Stage 5 Science
Integers are used to represent size and direction of physical quantities (vectors) such as displacement, velocity and acceleration.	Stage 6 Mathematics Advanced Stage 6 Mathematics Extension 1
Representation of integers on a number line is used to determine their absolute value.	Stage 6 Mathematics Advanced
Using customary or contemporary Aboriginal stories to create a number sentence to model the mathematical story such as ‘How many fish were caught in the fish traps and of these fish, how many fish were taken to eat?’	Aboriginal and Torres Strait Islander histories and cultures
Integers are used to show an increase or decrease in the amount of heat energy possessed by particles.	Stage 4 Science
Integers are used to compare positions above and below sea level, temperatures in different climates and life expectancy.	Stage 4/5 Geography
Integers are used to record time in years, changes in weights, results compared to records or personal bests, and in golf when comparing scores to par.	Real-world applications
Represents whole number integers on a number line, connects to Stage 3.	Stage 3 Represents Numbers B
Operations involving positive and negative integers are used in	Stage 4 Fractions,

comparing and ordering negative fractions.	Decimals and Percentages

Non-routine problem solving

- Find the largest possible number which can be made from 4 numbers using:
  - the 4 operations
  - the 4 operations and grouping symbols
- Give an example of an equation where the parentheses change the result of the computation.
- The product of 4 integers is 100. What are the possible sums of these 4 integers if they are:
  - all positive?
  - all negative?
  - either positive or negative?

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Fractions, decimals and percentages

Content

- Fractions, decimals and percentages

Teaching advice

Key ideas

Students:

- compare fractions using equivalence
- round decimals to a specified degree of accuracy
- identify terminating and recurring decimals
- define rational and irrational numbers
- order fractions, decimals and percentages and locate them on a number line
- convert between fractions, decimals and percentages
- represent operations with fractions
- express one quantity as a fraction, decimal or percentage of another
- solve problems involving the 4 operations on fractions and decimals
- solve problems involving percentages, including financial problems

Key terms

- fraction
- rational number
- improper fraction
- decimal
- highest common factor (HCF)
- percentage
- numerator
- recurring decimal
- number line
- denominator
- terminating decimal
- integer
- equivalent

Background knowledge

- The concepts of fractions are covered in Stages 1 to 3
- Order of operations
- Decimal place value
- Multiplication and division of decimals by powers of 10
- Benchmark fractions, decimals and percentages (10%, 25%, 50%, 75%, 100%)

Making connections

Connections	Outcome or Stage
Fractions, decimals and percentages are useful ways to express probabilities.	Stage 6 Num CEC 2.5 Stage 6 Num CEC 4.2.2 Stage 4 Probability Stage 5 Probability A and B Stage 6 Mathematics Standard Stage 6 Mathematics Advanced
Irrational numbers connect to Pythagoras' theorem. The number $\sqrt{2}$ arises as the hypotenuse of a right-angled triangle of side length 1.	Stage 4 Right-angled Triangles (Pythagoras' theorem)
The irrational number π is the ratio of circumference to diameter in a circle.	Stage 4 Length
Decimals are used in everyday life, for example fuel prices represented as 151.9c.	Stage 6 Num CEC 2.1.2
Fractions are used in everyday life, for example recipes.	Stage 6 Num CEC 2.1.1
Percentages are used in everyday life, for example when purchasing goods.	Stage 6 Num CEC 3.1
Percentages are linked to key ideas in finance, such as: ▪ GST, profit and loss, and best buys ▪ simple and compound interest, and depreciation ▪ dividend yield ▪ taxation	Stage 6 Num CEC 3.3.1 Stage 5 Financial Mathematics A and B Stage 6 Mathematics Standard
Error in measurements involves calculating with percentages.	Stage 5 Numbers of Any Magnitude Stage 6 Mathematics Standard
Percentage change can be calculated using statistical data.	Stage 6 Num CEC 4.2.1
Fractions are used in reading and interpreting musical note values such as semiquavers, quavers, crotchets, minims and semibreves. Musicians indicate fraction values by tails on the stems of notes or by	Stage 4 Music

contrasting open and closed notes.	
Identifying the percentage of bush tucker or native plants used in food products.	Aboriginal and Torres Strait Islander histories and cultures
Fractions, decimals and percentages are used: when comparing time related to work, leisure and rest; in calculating the height/weight ratio when analysing body composition; in assessing the effect of exercise on the body by measuring the increase in pulse rate and body temperature.	Real-world applications
Rounding of decimals is important for the ordering of decimals and estimation.	Stage 6 Num CEC 2.1.2
Conversion between fractions and decimals.	Stage 6 Num CEC 2.1.3
Fractions are used with algebraic techniques.	Stage 5 (Core) Algebraic Techniques – A
Ratios can be expressed as fractions of a whole.	Stage 4 Ratios and rates
Decimals are important in understanding how to express numbers in scientific notation.	Stage 5 Numbers of Any Magnitude Stage 6 Mathematics Standard
Fractions are important in using formulae such as Blood Alcohol Content (BAC) and calculating required medication dosages using Fried's, Young's or Clark's formula.	Stage 6 Mathematics Standard
Compare common fractions with related denominators.	Stage 3 Representing Quantity Fractions – B
Use equivalence to add and subtract fractional quantities (related denominators).	Stage 3 Representing Quantity Fractions – B

Non-routine problem solving

- Can a unit fraction always be written as the sum of 2 different unit fractions? Which unit fractions can be written as the sum of 2 different unit fractions in more than one way? Which unit fractions can only be written in one way?
- A certain fraction is greater than $1\frac{1}{3}$ and less than $1\frac{3}{5}$. If the denominator of this fraction is 30, how many possible values are there for the numerators?
- If $\frac{1}{6} = \frac{1}{A} + \frac{1}{B}$ where A and B are different whole numbers, find 4 different values of A and B.

- Consider the side length of a square whose area is 2 square units. Between which 2 whole numbers does the side length lie? Between which 2 decimal numbers (to one decimal place) does the side length lie? Show the position of the side length on a number line. Approximate the side length using digital tools.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Ratios and rates

Content

- Ratios and rates

Teaching advice

Key ideas

Students:

- explain the difference between a ratio and a rate
- simplify ratios and rates
- solve problems involving ratio, including using the unitary method
- divide quantities in a given ratio
- convert between rates
- solve rate problems including best buys
- interpret and construct distance–time graphs

Key terms

- rate
- ratio

Background knowledge

- Conversion of units of measurement and time
- Simplifying fractions using HCF

Making connections

Connections	Outcome or Stage
π is the ratio of circumference to diameter in a circle.	Stage 4 Length
Solving word problems involving rates connects to multiplication and division.	Stage 3 Multiplicative relations B
Ratios are used in the application of scales and maps.	Stage 5 Properties of geometrical figures A Stage 6 Num CEC 4.1.2 Stage 6 Mathematics Standard Stage 5 Geography
Scales are used to describe differences in size of and distance between structures making up the universe.	Stage 4 Science
Ratios are used when looking at enlargements and reductions in artworks and photographs and in copying pictures using scale grids.	Stage 4 Visual Arts Stage 4 Visual Arts (Ceramics) Stage 5 Visual Design Stage 5 Photography
Ratios are used in the mixing of agricultural chemicals, herbicides, sprays and drenches.	Stage 6 Num CEC 4.1.2
Rates are used in unit pricing, which is important in purchasing decisions, eg when comparing the cost-effectiveness of different options.	Stage 6 Num CEC 4.1.1
Rates are used to describe motion and calculating speed, distance and time.	Stage 6 Mathematics Standard
Ratio is connected to customary and contemporary Aboriginal and Torres Strait Islander maps, artworks or artefacts, eg identifying the scale used to represent real-life places.	Aboriginal and Torres Strait Islander histories and cultures
Ratios can be expressed as fractions of a whole.	Stage 4 Fractions, decimals and percentages
Ratios are important in Trigonometry such as trigonometric ratios of sine, cosine and tangent.	Stage 5 Trigonometry A Stage 6 Mathematics Standard Stage 6 Mathematics Advanced
Ratios are needed to solve problems involving scale factors in	Stage 5 Properties of

similar figures.	geometrical figures A Stage 6 Mathematics Standard
Distance–time graphs connect to linear relationships in real-life contexts.	Stage 4 Linear relationships
Distance–time graphs connect to rates of change.	Stage 5 Variation and rates of change B Stage 6 Mathematics Advanced Stage 6 Mathematics Extension

Non-routine problem solving

- A person takes 3 hours to paint a room and another person takes 5 hours to paint a room. How long will it take them to paint a room if they work together?
- Present a case for speed changes in wet conditions.
- I have 2 tins of paint in different shades of green. Both tins are exactly the same size. The first is a mix of green and white paint in the ratio 1:3. The second is a mix of green and white paint in the ratio 1:7. What is the ratio of green to white paint if we combine the 2 tins of paint?

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Teaching advice for Algebraic techniques

Content

- Algebraic techniques

Teaching advice

Key ideas

Students:

- use pronumerals to represent numbers
- apply the associative, commutative and distributive laws to algebraic expressions
- create algebraic expressions
- substitute into algebraic expressions
- generate number patterns from algebraic expressions
- identify like terms and simplify algebraic expressions involving the 4 operations
- expand and simplify algebraic expressions
- factorise algebraic expressions

Key terms

- operation
- commutative property
- factor
- factorise
- pronumeral
- pattern
- algebraic expression
- associative law
- distributive law

Background knowledge

- Order of operations
- Continuing a number pattern
- Multiplicative relations

Making connections

Connections	Outcome or Stage
The associative, commutative and distributive laws used in the arithmetic of numbers are applicable to algebra in the same way.	Stage 2 Multiplicative relations A and B Stage 3 Multiplicative relations A and B Stage 4 Computation with integers
Factorise numbers to aid mental multiplication.	Stage 3 Multiplicative relations A
Determine the highest common factor (HCF) of 2 whole numbers.	Stage 4 Fractions, decimals and percentages
Algebraic expressions will involve directed numbers.	Stage 4 Computation with integers
Algebraic expressions are used in area and volume calculations.	Stage 4 Area Stage 4 Volume
The generation of a number pattern from an algebraic expression leads into graphing.	Stage 4 Linear relationships
Simplifying algebraic expressions is used when solving equations.	Stage 4 Equations Stage 5 Equations
In spreadsheets, cell references act as pronumerals in formulae.	Real-world applications

Non-routine problem solving

- If a term has 3, 7, x and y as its factors find as many other factors of this term as possible.
- A student was asked to simplify $4y + 8y + 2$. Their answer was $14y$. How would you prove to the student that their answer is wrong?
- Write down a 2-digit number, then reverse the digits and add the 2 numbers. Compare your results with others in the class. Explain why the only possible sums are multiples of 11 and why the result is always 11 times the sum of the 2 digits.

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Teaching advice for Indices

Content

- Indices

Teaching advice

Key ideas

Students:

- interpret and express whole numbers in index notation with positive-integer indices
- identify and apply the basic index laws, including the zero index, with numerical examples
- calculate and simplify numerical expressions involving squares, cubes, square roots and cube roots
- identify the inverse relationship between square and square root, and cube and cube root

Key terms

- exponential
- indices
- index
- square numbers
- square root
- base
- cube root
- prime factors

Background knowledge

- Integers and arithmetic operations
- Order of operations
- Factors, division and multiplication
- Prime numbers
- Inverse operations

Making connections

Connections	Outcome or Stage
Expressing numbers in index notation with prime factors connects to factorisation, which will be extended into an algebraic context.	Stage 4 Algebraic techniques Stage 5 Algebraic techniques B Stage 5 Algebraic techniques C
Square and cube numbers connect to measurement in applications to area and volume.	Stage 4 Area Stage 4 Volume
Indices are used to represent very large and very small numbers such as distances in space and microscopic information about cells and blood.	Stage 5 Numbers of any Magnitude Stage 4 Science
Using indices to represent the distance between the Emu constellation and Earth at different times of the year, which is used to measure the time of the year in terms of seasons.	Aboriginal and Torres Strait Islander histories and cultures
Square roots are used to represent surds.	Stage 5 Indices C
Natural phenomena such as the pH scale and Richter scale for earthquakes involve properties of indices.	Stage 5 Science Stage 5 Logarithms
Exponential growth is represented using indices.	Stage 5 Non-linear relationships A
Model square numbers and record in numerical and diagrammatic form.	Stage 2 Multiplicative relations

Non-routine problem solving

- Present the legendary story of Sissa Ben Dahir who invented the game of chess. The king of India was impressed and granted him any reward. Sissa Ben Dahir chose a seemingly foolish reward: one grain of rice on the first square of the chessboard, then 2 on the second square, then 4 on the third square, then 8 on the fourth square and so forth. The king granted him his request … but who was the fool in the end?
- The following pattern uses prime factorisation:

$12 = 2^2 \times 3$

$12^2 = 2^4 \times 3^2$

$12^3 = 2^6 \times 3^3$

Use the pattern to represent the prime factorisation of 12^4 and 12^{10}. Find a general rule for the prime factorisation of 12^n.

- Using the digits 1 to 9, at most once each, create an expression of the form $\sqrt{\square\square\square} - \sqrt{\square\square} - \sqrt{\square}$ with a value as close to zero as possible.
- Explain how you can use prime factorisation to find the HCF of any 2 numbers.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Equations

Content

- Equations

Teaching advice

Key ideas

Students:

- recognise the difference between algebraic expressions and equations
- solve linear equations involving up to 2 steps
- solve simple quadratic equations of the form $ax^2 = c$
- translate word problems into equations and solve
- solve equations arising from substitution into formulas
- verify solutions by substitution

Key terms

- linear equation
- inverse operation
- quadratic equation
- operation
- equation
- algebraic expression
- square root

Background knowledge

- Inverse operations
- Simplifying algebraic expressions
- Substituting into algebraic expressions
- Square numbers and square roots

Making connections

Connections	Outcome or Stage
Completing number sentences that involve more than one operation by calculating missing numbers connects to equations.	Stage 3 Multiplicative relations B
Identifying and using inverse operations to assist with the solution of number sentences connects to equations.	Stage 3 Multiplicative relations B
Solving equations involves inverse operations and basic algebra skills.	Stage 4 Algebraic Techniques
Solving linear equations can involve applying graphs of linear relationships.	Stage 4 Linear relationships
Solving simple quadratic equations connects to square numbers and negative numbers.	Stage 4 Indices
Substitution into formulas is used to calculate area and volume. Conversely, the area model can be used to understand quadratics.	Stage 4 Area Stage 4 Volume
Solving simple quadratic equations is used when applying Pythagoras' Theorem.	Stage 4 Right-angled Triangles (Pythagoras' theorem)
Quadratic equations of the form $ax^2 = c$ connect to non-linear graphs (parabolas).	Stage 5 Non-Linear Relationships A
Formulas are used in spreadsheets.	Computing Technology 7–10 Real-world applications
Formulas are used to represent real-life phenomena such as density, force and temperature conversion.	Stage 4 Science
Formulas are used in financial applications such as simple interest, compound interest, depreciation and taxation.	Stage 5 Financial Mathematics A and B
Blood Alcohol Content (BAC) calculations involve formulas.	Stage 6 Mathematics Standard Stage 5 PDHPE

Non-routine problem solving

- Create a variety of equations that have the solution $x = 6$.
- Using integers, find an equation of the form $\frac{\square}{\square}x + \square = \square x + \frac{\square}{\square}$ which has (i) one solution, (ii) no solutions, (iii) infinitely many solutions.

- Think of a number. Multiply by 3. Add 5. Subtract the number you first thought of. Subtract 3. Divide by 2. Subtract 1. Explain why the answer is the number that you first thought of.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Linear Relationships

Content

- Linear relationships

Teaching advice

Key ideas

Students:

- identify and plot points on the Cartesian plane
- represent geometric patterns and number patterns using tables of values
- graph a line on a Cartesian plane using a table of values and graphing applications
- graph the points from the table of values on a number plane and join the points to demonstrate a linear relationship
- move flexibly between different forms of a linear relationship
- generate an algebraic rule to describe the number pattern observed
- solve linear equations by applying graphs of linear relationships using graphing applications
- find the point of intersection of 2 intersecting lines

Key terms

- Cartesian plane
- gradient
- linear relationship
- non-linear
- coefficient
- point
- constant
- coordinates
- intercept

Background knowledge

- Simple number patterns
- Creating algebraic expressions
- Substitution into algebraic expressions and basic formulas
- Using pronumerals to represent numbers
- Solving simple linear equations

Making connections

Connections	Outcome or Stage
Using and generating algebraic equations to represent patterns connects to basic algebra skills.	Stage 4 Algebraic Techniques
Using graphs to solve linear equations connects to solving simple equations algebraically.	Stage 4 Equations
Linear relationships are related to the concept of change, for example to time and distance for travel and migrations and changes in populations.	Stage 4 Ratios and Rates Stage 5 Variation and rates of change A and B Stages 4/5 Geography
Recording experimental and researched data in tables of values and presenting them as graphs involves linear relationships.	Stage 4 Data Analysis Stage 5 Data Analysis C Stages 4/5 Science Stages 4/5 Geography
Use a given geometric pattern involving multiples to create a table of values. This links through to sequences and series in Stage 6.	Stage 3 Multiplicative relations B Stage 6 Mathematics Advanced
Describing the relationship between the bottom number and the top number in a table links through to arithmetic sequences and series in Stage 6.	Stage 3 Multiplicative relations B Stage 6 Mathematics Advanced
Use the 4 quadrants of the coordinate plane.	Stage 3 Geometric measure B

Non-routine problem solving

- Conduct pattern investigations relating number planes and nature, eg Fibonacci sequence on bee colonies, sunflowers, spiral shells and the ‘golden ratio’.
- Use digital tools to determine and explain the differences between equations that represent linear relationships and those that represent non-linear relationships.
- If the sum of 2 numbers is 19 and their difference is 5, find the 2 numbers graphically.
- ‘All straight-line graphs are linear functions.’ Is this statement true or false? Justify your answer.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Length

Content

- Length

Teaching advice

Key ideas

Students:

- solve problems by finding the perimeter of plane shapes and composite figures involving quadrilaterals
- identify parts and features of a circle
- find the relationship between the diameter and circumference of a circle
- use the formulas $C = \pi d$ and $C = 2\pi r$ to solve problems involving circumference
- use the formula $l = \frac{\theta}{360} \times 2\pi r$ to solve problems involving arc length
- solve problems by finding the perimeter of sectors and composite figures involving sectors

Key terms

- perimeter
- circumference
- radius
- rhombus
- sector
- chord
- kite
- composite
- arc
- tangent
- pi
- angle
- segment
- approximate
- ratio
- subtend
- parallelogram
- circle
- diameter
- trapezium

Background knowledge

- Perimeters
- Side properties of special quadrilaterals
- Ratios
- Comparing part to whole in a fraction
- Calculating fractions
- Angle sum of a revolution is 360°
- Substitution into formulas
- Solving one-step linear equations
- Rounding numbers to specified number of decimal places

- Conversion between metric units

Making connections

Connections	Outcome or Stage
Arc lengths and perimeter of sectors directly connect to fractions, and eventually radian conversions.	Stage 4 Fractions, Decimals and Percentages Stage 6 Mathematics Advanced
Measure lengths to find perimeters.	Stage 3 Geometric measure A
Finding radii given circumferences or arc lengths connects to simple linear equations.	Stage 4 Equations
The concept of irrational numbers is also covered in fractions and indices (path).	Stage 4 Fractions, Decimals and Percentages Stage 5 Indices C
The relationship between the radius (or diameter) and circumference of a circle connects to the variation in sounds produced in different-sized Aboriginal and Torres Strait Islander instruments such as clap sticks, yidaki or the warup.	Aboriginal and Torres Strait Islander histories and cultures
Perimeters of plane shapes may involve the use of Pythagoras' theorem.	Stage 4 Right-angled Triangles (Pythagoras' theorem)
Calculation of perimeters of sporting fields, recreation parks, yarning circle spaces and farms involves selecting and using appropriate equipment, measuring tools and methods to make accurate observations and measurement, recording observations and measurements, and using appropriate units and abbreviations.	Stage 3 Geometric measure A Stage 6 Num CEC 1.3 Stage 6 Num CEC 1.3 Stage 6 Num CEC 2.4
Distances between 2 points on the equator can be found through calculations of arc lengths, involving latitude and longitude, and knowledge of the radius of the Earth.	Stage 6 Mathematics Standard 2
Fieldwork involves observing, measuring, collecting and recording information outside the classroom. Students use fieldwork instruments such as measuring devices, maps, photographs, compasses, GPS.	Stage 4/5 Geography

Non-routine problem solving

- Compare the perimeter of a regular hexagon inscribed in a circle with the circle's circumference to demonstrate that $\pi > 3$.
- Determine the difference in the distance travelled by a particle on the end of the hour hand, and one on the end of the minute hand of an analog clock.
- Examine the idea of an imaginary belt around the equator. This belt would touch the Earth at all points on its circumference (assuming the Earth was a perfect sphere or ball and had a radius of 6400 km). One metre is added to the length of the belt which would result in a small gap between the Earth and the belt. How wide is the gap between the Earth and the belt?
- Some tennis balls are sold in cylinders of 3 balls. Will the height of a cylinder that holds 3 tennis balls be greater than or less than its circumference? Justify your answer.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Right-angled triangles (Pythagoras' theorem)

Content

- Right-angled triangles (Pythagoras' theorem)

Teaching advice

Key ideas

Students:

- identify the hypotenuse in a right-angled triangle
- define Pythagoras' theorem
- apply Pythagoras' theorem to find the unknown length of a side of a right-angled triangle
- apply Pythagoras' theorem to solve practical problems
- apply the converse of Pythagoras' theorem to test whether a triangle is right-angled
- identify Pythagorean triads

Key terms

- hypotenuse
- square root
- Pythagoras' theorem
- Right-angle
- approximate
- converse
- square numbers

Background knowledge

- Substitution into algebraic expressions and formulas
- Squares and square roots of numbers
- Solving simple quadratic equations in the form $x^2 = c$
- Order of operations and addressing common misunderstandings (eg $3^2 + 4^2 \neq (3 + 4)^2$ and $\sqrt{5^2 - 4^2} \neq 5 - 4$)
- Rounding numbers to a specified degree of accuracy

Making connections

Connections	Outcome or Stage
Pythagoras' theorem connects to perimeters and areas involving plane shapes.	Stage 4 Length Stage 4 Area Stage 6 Mathematics Standard 2
The length of an interval joining 2 points on the Cartesian plane and the distance formula are derived using Pythagoras' theorem.	Stage 5 Linear Relationships A
Pythagoras' theorem is used to establish the equation of a circle with centre as the origin, the radius r and the graph of equations of the form $x^2 + y^2 = r^2$.	Stage 5 Paths Non-Linear Relationships C Stage 6 Mathematics Advanced
Pythagoras' theorem may be used in solving practical problems involving right-angled triangles and trigonometry.	Stage 5 Trigonometry Stage 6 Mathematics Standard 2 Stage 6 Mathematics Advanced
Connection to common content between Mathematics Standard 2 and Mathematics Advanced to solve practical trigonometry problems involving right-angled and non-right-angled triangles.	Stage 6 Mathematics Standard 2 Stage 6 Mathematics Advanced
Pythagoras' theorem is used in architecture and building such as roof pitch and pyramids.	Stage 6 Engineering Studies
Pythagoras' theorem is used in navigation and the flight paths of aircraft and mapping the navigation of ships	Stage 5 Trigonometry B
Right-angled triangles are important in sewing or quilting and in the use of timber.	Stage 4 Technology

Non-routine problem solving

- Does Pythagoras' theorem only work with squares? Explore what happens when we replace squares with semicircles. Do any other shapes work? Why/why not?
- Apply Pythagoras' theorem to find the height of an equilateral triangle and then derive the formula for the area of any equilateral triangle.
- Use Pythagoras' theorem to find the straight-line distance between 2 locations. Compare this to the distance travelled by road. Will it always be shorter?
- A right-angled isosceles triangle has a square drawn along each of its sides. The sum of the areas of the squares is 50 cm^2. What is the area of the triangle?

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Area

Content

- Area

Teaching advice

Key ideas

Students:

- apply formulas to find the area of triangles, rectangles, parallelograms, trapeziums, rhombuses, kites, circles and sectors
- find the areas of composite shapes consisting of triangles, rectangles and parallelograms
- find the areas of sectors and related composite shapes
- use and convert between appropriate units of measurement

Key terms

- quadrilateral
- triangle
- sector
- circle
- parallelogram
- hectare
- trapezium
- kite
- rectangle
- rhombus

Background knowledge

- Combine, split and rearrange shapes to determine the area of parallelograms and triangles
- Using metric units for length
- Ratios
- Comparing part to whole in a fraction
- Calculating a fraction of a quantity
- Angle sum of a revolution is 360°
- Substitution into formulas
- Rounding decimals
- Solving linear equations

Making connections

Connections	Outcome or Stage
Find the area of composite figures (rectangles).	Stage 3 Two-dimensional spatial structure B
Calculate the area of a parallelogram using subdivision and rearrangement.	Stage 3 Two-dimensional spatial structure B
Determine the area of a triangle.	Stage 3 Two-dimensional spatial structure B
Using multiplication of distances and dividing composite shapes into known shapes.	Stage 6 Num CEC 1.3
Circular areas may connect to gathering spaces in Aboriginal and Torres Strait Islander cultures.	Aboriginal and Torres Strait Islander histories and cultures
Calculating areas of sectors connects to fractions or parts of a whole and sector graphs.	Stage 4 Data Classification and Visualisation Stage 4 Fractions, Decimals and Percentages
Calculating areas of shapes is used in finding surface areas and volumes of solids.	Stage 4 Area Stage 4 Volume
Connecting rates to land areas to determine population density.	Stage 5 Geography Stage 4 Ratios and Rates
Strong connections to agriculture include calculating field area including irrigated circular crops as well as broadacre crops, conversion of area units, land area and application of fertiliser and herbicides, density of stock, or plants for farming.	Real-world applications
Connections to area exist in graphic design and works of art in terms of paper sizes.	Stage 4 Visual Arts

Non-routine problem solving

- Two triangles have an area of 18 cm^2. Will they have the same perimeter? Justify your answer.
- Compare various shapes with the same perimeter and ask questions related to their area such as whether they have the same area. What would be the maximum possible area for a given perimeter? What is the largest area for a given perimeter?

- Create your name using exactly 100 square centimetres and at least one of each shape: rectangle, trapezium, rhombus, triangle, kite and parallelogram. Why can't you include circles in this activity?

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Volume

Content

- Volume

Teaching advice

Key ideas

Students:

- identify, describe and represent different views of a right prism
- apply the formula to find the volume of a right prism
- apply the formula to find the volume of a cylinder
- choose appropriate units for volume and capacity and convert between them
- identify the difference between volume and capacity and solve related problems

Key terms

- capacity
- perpendicular
- cross-section
- prism
- cylinder
- volume

Background knowledge

- Visualisation, sketching and construction of three-dimensional objects, including prisms and pyramids
- Estimation and measurement of volumes and capacities
- Substitution into a given formula
- Metric units of length and area
- Prefixes such as kilo and mega in real-world contexts
- Calculation of the area of rectangles, triangles and circles

Making connections

Connections	Outcome or Stage
Use of formula links to equations and solving one- and two-step equations.	Stage 4 Equations Stage 5 Equations A
Connect three-dimensional objects with two-dimensional representations.	Stage 3 Three-dimensional spatial structure A
Recognise the multiplicative structure for finding volume.	Stage 3 Three-dimensional spatial structure B
Find the volumes of rectangular prisms in cubic centimetres and cubic metres.	Stage 3 Three-dimensional spatial structure B
Volume connects to the Brewarrina fish traps and the changing volume of water inside the rectangular traps at various times throughout the year.	Aboriginal and Torres Strait Islander histories and cultures
Many practical volume problems connects to rates such as water flow.	Stage 4 Ratios and Rates Stage 6 Mathematics Standard 1
Links to topographical maps in Geography – the shape of flood plains affects the impact of flooding, including the rate that water rises and falls.	Stage 4/5 Geography
Plan view of 3D shapes links to house plans and construction, including collection of rainfall runoff.	Stage 6 Num CEC 3.5 Stage 6 Mathematics Standard 1
Volume calculations in everyday situations.	Stage 6 Num CEC 1.3 Technology 7–8
Capacity calculations in everyday situations.	Stage 6 Num CEC 2.4

Non-routine problem solving

- Compare a rectangular prism and a cylinder. Work out the volume of the rectangular prism and compare it to a cylinder with the same height. Compare the surface areas of the 2 prisms to compare maximum volume to minimum surface area. Recognise that prisms with the same volume may have different surface areas, and prisms with the same surface area may have different volumes.
- Create a new package that minimises the surface area while maintaining the volume.
- I have a box that is a rectangular prism. It has a volume of 60 cm^3. What are the dimensions of my box? What happens to the volume of my box if I double its dimensions?

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Angle relationships

Content

- Angle relationships

Teaching advice

Key ideas

Students:

- use relevant notations, language and conventions for angle relationships, including parallel and perpendicular lines
- describe and identify geometrical properties for angles at a point
- apply properties of parallel lines cut by a transversal to solve problems
- determine and justify whether 2 lines are parallel
- use given information to find sizes of unknown angles in a visual representation

Key terms

- point
- transversal
- supplementary
- reflex angle
- line
- parallel
- adjacent angles
- vertically opposite angles
- ray
- perpendicular
- right angle
- alternate angle
- interval
- vertex
- straight angle
- corresponding angles
- angle
- complementary angles
- revolution
- co-interior angles

Background knowledge

- Identification and classification of angles by comparing to a right angle
- Measurement and construction of angles, and identification of the relationships between angles on a straight line and angles at a point

Making connections

Connections	Outcome or Stage
Analysing the relationships formed by the intersection of straight lines.	Stage 3 Geometric Measure B
Properties of angles in parallel lines are used in trigonometry in relation to angles of depression and bearings.	Stage 5 Trigonometry B
Geometrical properties of angles at a point are used in art, building and construction, and tessellation using geometric shapes.	Stage 4 Properties of Geometrical Figures Stage 4 Visual Arts
Angles are used by Aboriginal Peoples and Torres Strait Islander Peoples in navigation, using both landmarks and celestial navigation.	Aboriginal and Torres Strait Islander histories and cultures
Aboriginal Peoples' customary dwellings use various design elements relating to particular angles and geometric figures needed to construct dwellings that protect from the elements.	Aboriginal and Torres Strait Islander histories and cultures
Deductive reasoning is used in proofs involving numerical, algebraic and trigonometric identities.	Stage 5 Properties of Geometrical Figures A
Relevant notations and conventions and angle relationships are used in circle geometry and vectors.	Stage 5 Circle Geometry Stage 6 Mathematics Extension 1 Stage 6 Mathematics Extension 2
Angles associated with parallel lines are involved in many scientific concepts including collisions, reflection and refraction.	Stage 6 Physics
Angles are considered when teaching students to make correct shapes with their bodies in Dance Performance and when composing shapes in Dance Composition.	Stage 4 Dance

Non-routine problem solving

- Use the properties of parallel lines and transversals to explain how ironing boards always remain horizontal.
- Examine Eratosthenes' calculation of the circumference of the Earth using parallel line results.
- Determine the properties required for a shape to tessellate. Test your theory by creating an artwork from an irregular shape.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Properties of geometrical figures

Content

- Properties of geometrical figures

Teaching advice

Key ideas

Students:

- use relevant notations and conventions for naming and labelling triangles and quadrilaterals
- classify, describe and use properties of triangles and quadrilaterals
- prove and use the angle sum of a triangle and quadrilateral
- solve numerical problems using properties of triangles and quadrilaterals

Key terms

- quadrilateral
- obtuse
- rhombus
- side
- equilateral triangle
- parallelogram
- kite
- length
- isosceles triangle
- rectangle
- trapezium
- acute angle
- scalene triangle
- square
- right angle
- triangle
- interior angle

Background knowledge

- Basic side and angle properties of triangles and quadrilaterals using measurement and symmetry
- Angle relationships
- Solving equations

Making connections

Connections	Outcome or Stage
Geometric shapes are used in design, architecture and building.	Stage 6 Num CEC 3.5
Classify two-dimensional shapes and describe their properties.	Stage 3 Two-dimensional spatial structure A
Deductive reasoning is used in proofs involving numerical, algebraic and trigonometric identities.	Stage 5 Properties of Geometrical Figures A
Properties of triangles and quadrilaterals are assumed knowledge for Mathematics Advanced optimisation problems (applications of the derivative).	Stage 6 Mathematics Advanced
Properties of triangles and quadrilaterals are used in vector manipulation and vectors are used to prove geometric results.	Stage 6 Mathematics Extension 1 Stage 6 Mathematics Extension 2

Non-routine problem solving

- A cantilevered design is commonly used in toolboxes, fishing boxes and sewing boxes. Use the properties of parallelograms to explain why the trays of these boxes stay horizontal as they are opened, so that the contents do not spill.
- Prove the geometrical properties of triangles and quadrilaterals using coordinate geometry, with and without the use of digital tools.
- Create a hierarchy of quadrilaterals based on their properties.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Data classification and visualisation

Content

- Data classification and visualisation

Teaching advice

Key ideas

Students:

- classify data as numerical (discrete or continuous) or categorical (nominal or ordinal)
- display data using appropriate graphical representations including frequency histograms and polygons, dot plots, stem-and-leaf plots, divided bar graphs, line graphs, sector graphs and pictograms
- interpret a wide variety of graphical representations and develop skills in graph comprehension
- represent a single dataset using an infographic
- identify and describe how graphical representations can be misleading

Key terms

- data
- numerical data
- infographic
- discrete variable
- categorical variable
- continuous variable
- ordinal
- graph
- stem-and-leaf plot

Background knowledge

- Construction of graphs and their features
- Ability to work with simple datasets and create simple graphical representations
- Understanding of scale and features of graph
- Angle sum of a revolution is 360°

Making connections

Connections	Outcome or Stage
Statistical examination of a particular 'variable' (a value or characteristic that changes for different individuals) is of fundamental importance and is used widely in everyday situations in fields including education, business, economics and government.	Stage 6 Num CEC 4.2.1 Real-world applications
Interpret and compare a range of data displays.	Stage 3 Data B
Interpret data presented in digital media and elsewhere.	Stage 3 Data B
Data can be used to represent a range of information about Aboriginal and Torres Strait Islander Peoples, histories and cultures, such as maintenance and revival of languages over time, success of community-controlled health initiatives, Aboriginal-owned enterprises.	Aboriginal and Torres Strait Islander histories and cultures
ABS population data related to Aboriginal Peoples may be connected to the size of circles in Aboriginal artworks that represent the relative size of local clan groups. The ratio of circles may also be used.	Aboriginal and Torres Strait Islander histories and cultures
Graphical representations of data are used to process and analyse data and information. Students may carry out investigations to test or research a problem or hypothesis and then record and analyse the data and identify trends, patterns and relationships.	Stages 4, 5 and 6 Geography Stages 4, 5 and 6 Science
Infographics, visual representations and various graphs are used in many other subjects, in particular Science and Geography.	Stage 4, 5 and 6 Science Stage 4, 5 and 6 Geography
Strong connections to Stage 5 Data Analysis A.	Stage 5 Data Analysis A

Non-routine problem solving

- Use different representations of the same dataset to promote a given point of view or provide a persuasive argument.
- Predict changes to visualisations by manipulating datasets to purposely skew the data. Examples include removing small or large data values at the boundaries and adding in outliers.
- Research Benford's Law and see how this distribution compares with real-life data such as house numbers of people in their class or year group, populations of cities in a country, lengths of longest rivers in the world or lead digits of several scientific constants.
- Carry out a simple survey of the class. Come up with 3 or 4 ways to display the data. What story does each display tell? When might one graphical representation be used in favour of another?

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Data analysis

Content

- Data analysis

Teaching advice

Key ideas

Students:

- calculate and compare the mean, median, mode and range for simple datasets
- determine the most appropriate measure of centre for a dataset in the context of the data
- identify and describe the effects of outliers on a dataset
- describe the effect of clusters, gaps and outliers on the distribution shape of a dataset
- analyse sets of data presented in a variety of ways
- identify symmetry and skewness in data displays
- discuss what conclusions can be drawn from the data presented and express the meaning of that information

Key terms

- data
- mode
- skewness
- average
- range
- measure of centre
- sample
- mean
- dataset
- distribution
- population
- median
- outliers
- symmetrical distribution

Background knowledge

- The term 'average', when used in everyday language, generally refers to the mean and describes a 'typical value' within a set of data
- Symmetry of shapes
- Representations of single datasets

Making connections

Connections	Outcome or Stage
Students could investigate real-world data such as COVID-19 vaccination rates, or objective data relating to sporting performance.	Stage 6 PDHPE Stage 5 Science Real-world applications
Interpret and compare different displays in terms of the distribution shape, including the range and mode.	Stage 3 Data B
Data literacy and representing data collected from experiments, such as chemical reactions and nutritional data is connected to this outcome. It is important to synthesise, analyse and make decisions and predictions.	Stages 4, 5 and 6 Science
Aboriginal and Torres Strait Islander Ranger groups collect and analyse data, such as to identify and monitor animal and plant populations and water quality.	Aboriginal and Torres Strait Islander histories and cultures
Understanding when best to use mean, mode or median and their differences is an important part of interpreting information in the media.	Stage 6 Num CEC 4.2.1
The skills to identify influences such as clusters, gaps and outliers contribute to analysis of accuracy and misleading graphical representations.	Stage 5 Data Analysis A
Shapes of data displays are analysed when considering the normal distribution and other probability and statistical distributions.	Stage 6 Mathematics Advanced Stage 6 Mathematics Extension 1
Analysing data is key to tracking climate change and providing information to the public.	Stage 6 Geography

Non-routine problem solving

- A student's test scores are 95, 82, 76 and 88. What score must they get on the fifth test in order to achieve an average of 84 on all 5 tests?
- Examine the use of stacked bar charts, area graphs and line graphs, discussing how they represent larger categories that can be subdivided into smaller categories and how information that can be obtained from these displays can be used for comparison.
- Examine the spread of salaries in a single sports team. How do salaries vary within the team? Compare the spread across different teams and across different sports.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Teaching advice for Probability

Content

- Probability

Teaching advice

Key ideas

Students:

- list the sample space of an event
- determine probabilities of single-step chance experiments as fractions, decimals or percentages
- recognise that the sum of the probabilities of all possible outcomes is 1
- explore relative frequencies by repeating a chance experiment a number of times using digital tools
- determine probabilities for complementary events
- compare theoretical and experimental probabilities using results from simulations

Key terms

- probability
- equally likely outcomes
- relative frequency
- sample space
- bias
- chance
- complementary events

Background knowledge

- The probability of an impossible event is 0, and a certain event has a probability of 1
- All probabilities lie on a scale from 0 to 1
- Conversions between fractions, decimals and percentages

Making connections

Connections	Outcome or Stage
Create random generators and describe probabilities using fractions.	Stage 3 Chance B
Conduct chance experiments with both small and large numbers of trials.	Stage 3 Chance B
Probabilities of single-step chance experiments form the basis for determining the probability of multistage events.	Stage 5 Probability A
Aboriginal and Torres Strait Islander Peoples use local animal and plant behaviours to predict the probability of events occurring, eg kookaburras calling during the day may indicate a high probability of rain.	Aboriginal and Torres Strait Islander histories and cultures
Experimental probability connects with relative frequency in statistics.	Stage 6 Mathematics Standard Stage 6 Mathematics Advanced
Probabilities connect to discrete and continuous random variables.	Stage 6 Mathematics Advanced Stage 6 Mathematics Extension
Probabilities connect to combinatorics and the binomial theorem.	Stage 6 Mathematics Extension
Probabilities are used when analysing risk for insurance calculations. For example, insurance premiums that are designed for young adult males who have a higher probability of accidents and insurance claims, and the likelihood of natural disasters such as flooding and bushfires which also affects insurance premiums.	Real-world applications
Card games and sports statistics can be used to predict the probability of a team winning or losing.	Stage 6 Num CEC 2.5 Stage 6 Num CEC 4.2.2

Non-routine problem solving

- Use the probability of elements to make decisions in activities based on game-show formats, such as *Deal or No Deal*. Consider strategies that bring the greatest ‘success’ based on probabilities that are dynamic throughout the game.
- In a classroom of 23 students, what is the probability that at least 2 share a birthday?
- Examine the law of small numbers and the gambler’s fallacy.

Further advice is given in support materials which provide teaching strategies, supports and scaffolds demonstrating how to support the diversity of learners in Mathematics 7–10.

Glossary

Mathematics K–10 (2022) Terms

absolute error

In a measurement, half of the smallest unit of the measuring device. The smallest unit is called the precision of the device.

access content points

Content for students with significant intellectual disability who are working towards the Early Stage 1 outcomes. Teachers can use the access content points on their own, or in combination with the content for each outcome.

accuracy

The extent to which measurements are considered to be close to the true value.

acute angle

An angle smaller than a right angle, between 0° and 90°.

Extended in Mathematics Advanced and Extension 1: In circular measure, an angle lying in the interval $\left[0, \frac{\pi}{2}\right]$.

adjacent

Has different meanings for different mathematical contexts. For example, adjacent vertices in a network are joined by an edge and adjacent faces in a solid have a common edge. In trigonometry, the adjacent side of a right-angled triangle is the side next to the angle to which the trigonometric ratio is referring.

adjacent angles

Angles that share a vertex and an arm. They do not overlap.

Extended in 7–10: Two angles at a point are called adjacent if they share a common ray and a common vertex and lie on opposite sides of the common ray.

algebraic expression

A mathematical statement formed by combining numbers and algebraic symbols using arithmetic operations, e.g. $a^2 + 3ab - 2b^2$.

algebraic fraction

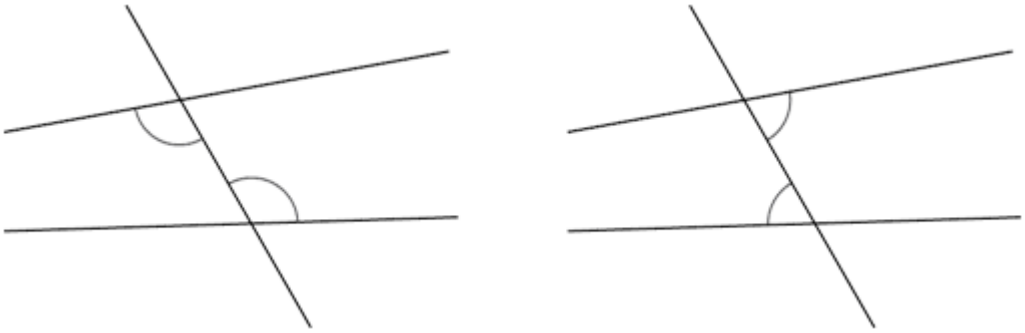
A fraction in which the numerator and/or the denominator are algebraic expressions.

algebraic term

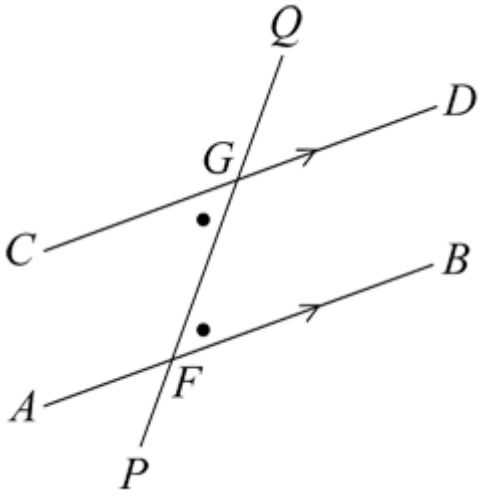
Forms a 'separable' part of some other algebraic expression. For example, x^2 and $5x^{-1}$ are terms in the inequality $x^2 \leq 5x^{-1}$, and 2, $-3x$, $5x^2$ are terms of the polynomial $2 - 3x + 5x^2$.

alternate angle

In each diagram below, the 2 marked angles are called alternate angles (since they are on alternate sides of the transversal).



If the lines AB and CD are parallel, then each pair of alternate angles is equal.

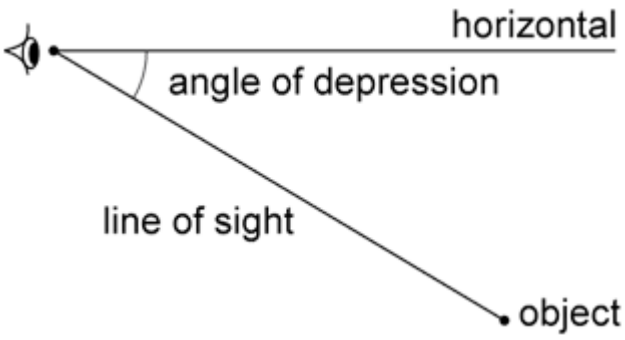


angle

Formed by 2 straight lines meeting at a common endpoint, called the vertex. An angle can describe the amount of turn between its 2 arms (lines).

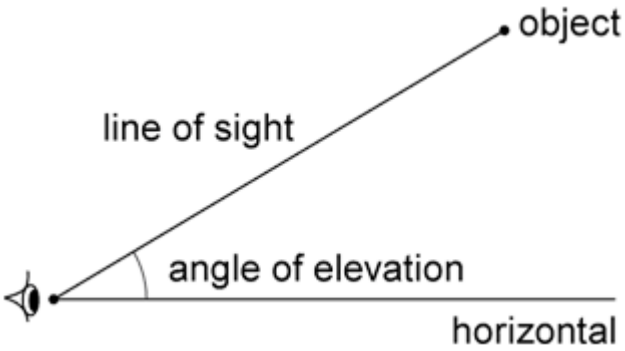
angle of depression

The angle between horizontal and the line of sight from an observer to an object that is lower than the observer.



angle of elevation

The angle between horizontal and the line of sight from an observer to an object that is higher than the observer.



angle of inclination

The angle a straight line makes with the positive x-axis.

approximate

An estimate of a number or an amount to a particular accuracy. For example, $\frac{22}{7}$ and 3.1416 are approximate values, or approximations, for the number π.

arc

A part of a circle's circumference.

Extended in Mathematics Extension 1: A part of a curve.

array

One of several different arrangements that can be used to model multiplicative situations. It is made by arranging a set of objects, such as counters, into columns and rows. Each column must contain the same number of objects as the other columns, and each row must contain the same number of objects as the other rows.

associative law

When more than 2 numbers are added or multiplied, the result is unchanged regardless of how they are grouped or associated. For example, addition of numbers is associative, and the corresponding associative law is: $(a + b) + c = a + (b + c)$ for all numbers a, b and c.

Multiplication is also associative: $(ab)c = a(bc)$ for all numbers a, b and c.

asymptote

A straight line (or another curve) that a curve approaches as x tends to $\pm\infty$, or to some particular value.
For example, the curve $y = 2^x$ has an asymptote $y = 0$ as x tends to $-\infty$, and the curve $y = \frac{1}{x}$ has asymptotes $x = 0$ and $y = 0$ as x tends to 0 and $\pm\infty$ respectively.

attribute

A quality, feature or characteristic of an object or shape.

average

A number expressing a central or typical value in a set of data. While it usually refers to the arithmetic mean, that is, the sum of a set of numbers divided by the number of numbers in the set, it may also refer to other measures of centre.

base

Has different meanings for different mathematical contexts. A base is a number that is raised to a power and the base of a solid is the face the object rests on.

bearing

A direction from one point on the Earth's surface to another. Two types of bearings may be used: compass bearing and true bearing.

See compass bearing and true bearing

bias

A systematic favouring of certain outcomes more than others, due to unfair influence (knowingly or otherwise).

binomial expression

An algebraic expression with 2 terms. For example, $x + y$ is a binomial expression in the 2 terms x and y.

bivariate data

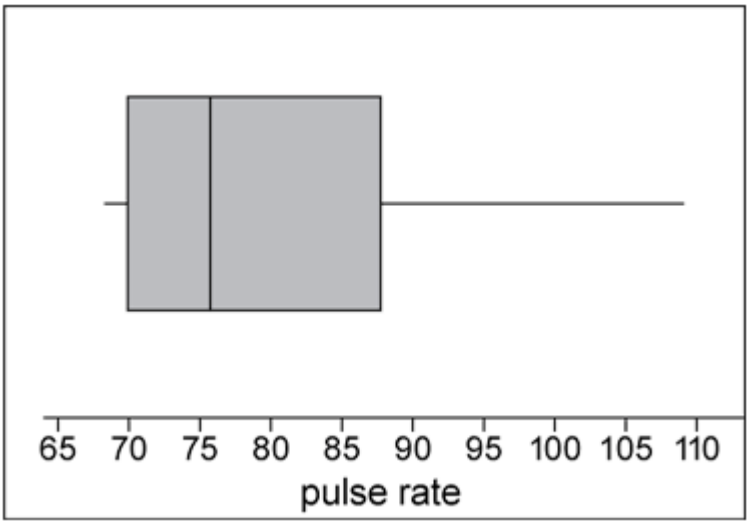
Data relating to measurement of 2 variables. It can be categorical data, numerical data or a combination of both.

box plot

A synonym for a box-and-whisker plot. A graphical display of a 5-number summary of data. The 'box' covers the middle 50% of scores. The 'whiskers' extend to the maximum and minimum values in the dataset.

In a box plot, the 'box' covers the interquartile range (IQR). A vertical line in the box is used to indicate the location of the median.

The box plot below has been constructed from the 5-number summary of the resting pulse rates of 17 students.



capacity (internal volume)

Generally refers to the amount a container can hold.

A term used to describe how much a container will hold. It is often used in relation to the volume of fluids. Units of capacity (volume of fluids or gases) include litres and millilitres.

See internal volume (capacity)

Cartesian coordinate system

A system that specifies the position of each point in a plane, formed by the intersection of 2 perpendicular number lines called axes. Each point is represented by an ordered pair $(x, y)$of real numbers called the coordinates of the point. The point where the axes meet is called the origin and has coordinates $(0,0)$.

categorical variable

A variable whose values belong to exactly one of a number of categories. A categorical variable describes a quality or characteristic of something. Sometimes called a discrete variable. There are 2 types of categorical variables: nominal and ordinal. For example, your home state or blood type are categorical variables.

causal

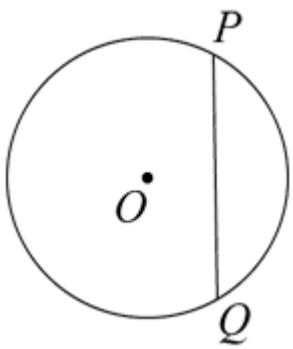
Two variables with a relationship which indicates that one event is the result of the occurrence of the other event.

chance

A possibility of something happening.

chord

A line segment (interval) joining 2 points on a circle.



circle

All points equal to a given distance from a fixed point, the centre. When used to describe a shape, a circle includes all points inside the boundary.

circuit (networks)

A path that begins and ends on the same vertex. Vertices can be repeated.

circumference

The boundary of a circle. The length of the circumference C is given by $C = \pi d$, where d is the diameter. Alternatively, it is given by $C = 2\pi r$, where r is the radius.

closed shape

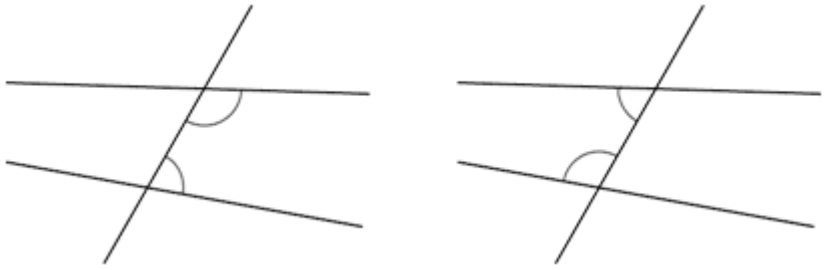
A shape enclosed by all its sides with no gaps.

co-interior angles

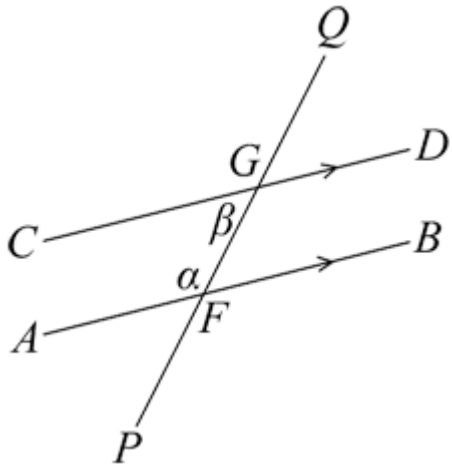
A pair of angles that lie between a pair of parallel lines cut by a transversal and have a sum of 180 degrees or 2 right angles.

In each diagram the 2 marked angles are called co-interior angles.

If the 2 lines are parallel, then co-interior angles add to give 180° and so are supplementary.



Co-interior angles formed by parallel lines are supplementary. In the diagram below the angles ∠CGF and ∠AFG are supplementary. If the lines AB and CD are parallel then $\alpha + \beta = 180$:



Conversely, if a pair of angles are supplementary, then the lines are parallel.

coefficient

A numerical quantity that multiplies a variable in an algebraic expression, e.g. 5 is the coefficient of $5x$. Variables with no specified coefficient have a coefficient of 1.

collinear

Describes points that lie on the same straight line.

combinations

The various ways a group of items can be arranged without regard to order.

commission

The sum or percentage paid for services, e.g. to an agent or salesperson.

common factor

A factor of each element of a set of numbers or an algebraic expression. Also known as common divisor. For example, 6 is a common factor of 24, 54 and 66, and $x + 1$ is a common factor of $x^2 - 1$ and $x^2 + 5x + 4$.

commutativity (commutative property)

Commutativity (commutative property) of addition or multiplication means that 2 numbers can be added or multiplied in any order and the solution will be the same.

Commutative law, commutativity and turn-around facts are interchangeable terms.

compass bearing

Angles either side of north or south. For example, a compass bearing of N50°E is found by facing north and moving through an angle of 50° to the east.

complementary angles

Two adjacent angles that form a right angle, i.e. the sum of the angles measured in degrees is 90°.

complementary events

Two mutually exclusive outcomes in a probability experiment, such that $P(A) + P(B) = 1$ where $P(A)$ is the probability of event A and $P(B)$ the probability of event B.

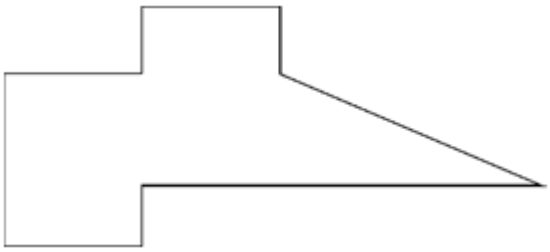
Extended in Mathematics Standard and Advanced: Two mutually exclusive events that make up the sample space.

composite number

A non-zero natural number that has a factor other than 1 and itself. For example, all even numbers besides 2 are composite numbers. Some odd numbers are composite (e.g. 21) but not all are composite (e.g. 11).

composite shape

A shape that is formed by combining other plane shapes. Composite shapes are often described as 'complex' when they are made up of many and different shapes.



compound interest

The interest earned by investing a sum of money (the principal) when each successive interest payment is added to the principal for calculating the next interest payment.

concavity

Indicates whether the curve ‘opens’ up (concave up) or ‘opens’ down (concave down). For example, the parabola $y = x^2$ is concave up and the parabola $y = -x^2$ is concave down.

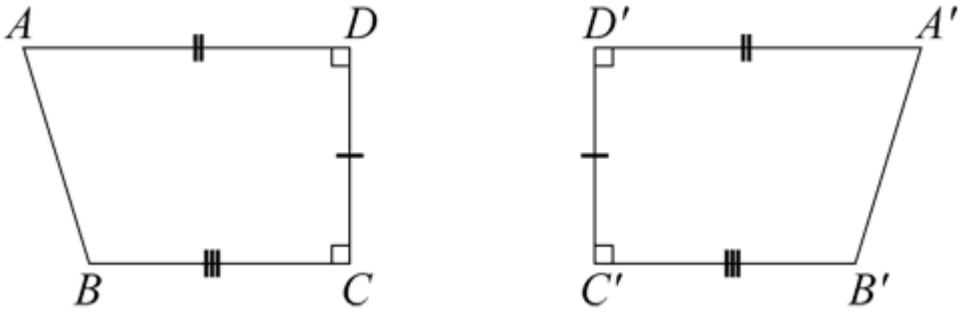
conditional probability

Consideration of whether the knowledge of the occurrence of one event, A, affects the probability of occurrence of another event, B.

Extended in Mathematics Advanced: The probability of an event A occurring, given that another event B has occurred, notated as $P(A|B)$ (read as ‘probability of A given B’).

congruent

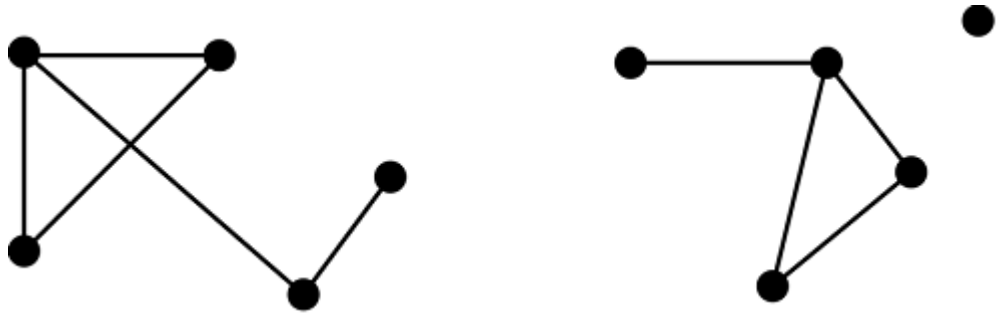
An exact match between every part of one figure with the corresponding part of another figure, i.e. congruent figures can be exactly superimposed on each other.



connected

In the Cartesian plane, 2 points are connected if there is a line or curve that joins them. In a network, a set of points is connected if every point can be reached from another point. A set of points that is not connected is called disconnected.

Examples of a connected network and a disconnected network respectively are shown below:



conserved

Not created or destroyed.

constant

A fixed numerical value. For example, in the algebraic expression $x + 11$, the number 11 is a constant.

constant difference

Constant difference refers to a common difference between pairs of numbers. For example, the difference between 3 and 7 is 4, and another pair of numbers that has a difference of 4 is 2 and 6.

container

Objects such as bottles, jars and tubs.

continuous variable

A numerical variable that can take any value that lies within an interval. The values taken are subject to the accuracy of the measuring instrument used to obtain these values. For example, height, reaction time to a stimulus and systolic blood pressure.

converse

If one relationship is true, then the opposite is also true.

The converse of a statement 'If P then Q' is 'If Q then P'. For example, the converse of Pythagoras' theorem is that if the sum of the squares of 2 sides of a triangle is equal to the square of the third side, then the triangle is right-angled.

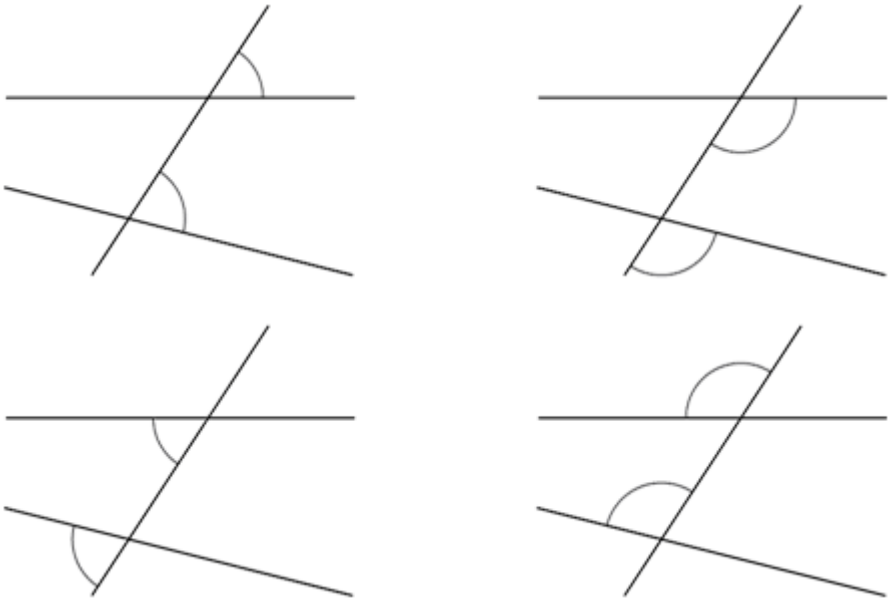
Extended in Mathematics Extension 2: The converse of $P \rightarrow Q$ is $Q \rightarrow P$ or $P \leftarrow Q$.

coordinates

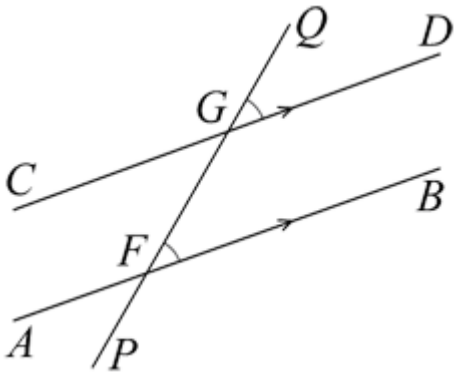
A set of values that define the position of a point or an object in a space. The coordinate system being used will govern the nature of the coordinates.

corresponding angles

Angles that lie in the same relative position at the points of intersection where a pair of lines is cut by a transversal.



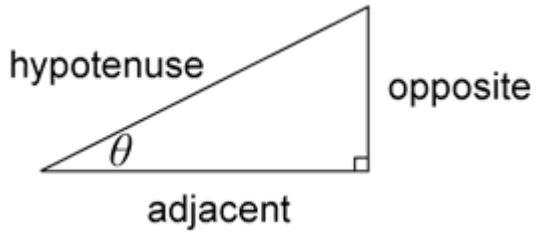
If the lines are parallel, then each pair of corresponding angles is equal.



Conversely, if a pair of corresponding angles is equal, then the lines are parallel.

cosine ratio

The ratio of the length of the base, b, of a right-angled triangle to the length of its hypotenuse, h, for a given base angle theta, θ. In any right-angled triangle, $\cos\theta = \frac{adjacent}{hypotenuse}$, where $0 < \theta < 90°$.



cross-section

The intersection of a solid with a plane. May differ for a given solid depending on the orientation of the slicing plane. For example, the cross-section of a sphere will be a circle (of different radius depending on the slicing plane), while the cross-section of a cube could be a square or other polygon (e.g. triangle, hexagon) depending on the orientation of the slicing plane.

cube

An object with 6 equal square faces and 12 edges.

cube root

The inverse or opposite of cubing a number. For example, the cube root of 8 is 2 because $2 \times 2 \times 2 = 8$. In symbols it is written $\sqrt[3]{8} = 2$.

cubic

Having the form of a cube or being of the third power or degree.

curve

A one-dimensional figure that is not straight.

curved surface

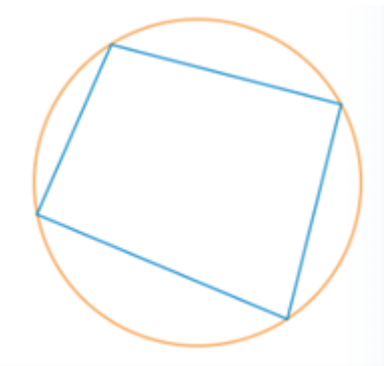
Surface of an object that is not flat, e.g. a cylinder has one curved surface.

cycle (networks)

A trail that finishes on the vertex it begins on. No vertex is repeated.

cyclic quadrilateral

A quadrilateral with every vertex (corner point) on a circle's circumference.



cylinder

An object that has parallel circular discs of equal radius at the ends that are joined by a curved surface.

data

Facts or units of information collected together.

dataset

A collection of numbers or values relating to a particular subject. Datasets are normally presented in tables or represented by graphs. For example, the test scores of each student in a particular class.

decimal

Used to describe aspects of the base-10 number system. The decimal point (. or ,) separates the whole number part of a number from its decimal part.

degree (networks)

The degree of a vertex is the number of edge ends at that vertex.

degrees (geometry)

A unit for measuring an angle. Angles are measured as a proportion of a full turn, which is equivalent to 360 degrees, so that one degree, written as $1°$ is equal to $\frac{1}{360}$ of a full turn.

denominator

Comes from Latin, meaning 'that which gives a name' [de- "completely"; nomen "name"]. The denominator of a fraction identifies the name of the fractional parts (e.g. thirds, quarters or fifths).

Extended in 7–10: In the fraction $\frac{a}{b}$, b is the denominator.

dependent events

An event that relies on another event to happen first. One event influences the probability of another event.

dependent variable

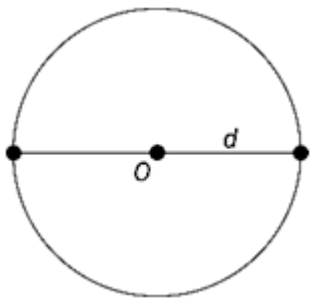
The variable used to represent the output values of a function. A dependent variable is generally represented on the vertical axis of a graph.

depreciation

A decrease in value due to wear and tear, decay, decline in price etc.

diameter

A chord passing through the centre of a circle. The word is also used for the length of the diameter.



difference

The difference of 2 numbers is the *result* of subtracting one from the other. You can use subtraction or addition to find the difference between 2 numbers.

For example, to find the difference between 15 and 21, you can:

- add 5 up from 15 to get to 20, then add one more to get to 21, so the difference is 6
- start with 21 and count back to find the difference of 6.

direct variation

A proportional relationship where one quantity directly varies with respect to a change in another quantity. This implies that if there is an increase (or decrease) in one quantity then the other quantity will experience a proportionate increase (or decrease).

discrete variable

Individual and countable items that can be listed.

Extended in Mathematics Advanced: Also known as a discrete random variable.

distribution

The spread of data which can be shown graphically or in a table.

distributivity (Distributive law)

Multiplication of numbers is distributive over addition because the product of one number with the sum of 2 others equals the sum of the products of the first number with each of the others. For example, the product of 3 with $(4 + 5)$ gives the same result as the sum of 3×4 and 3×5:

$3 \times (4 + 5) = 3 \times 9 = 27$ and $3 \times 4 + 3 \times 5 = 12 + 15 = 27$

The distributive law is expressed algebraically as follows:

$a(b + c) = ab + ac$, for all numbers a, b and c.

dividend

The number that is divided or distributed in a division problem. For example, in $12 \div 3 = 4$, 12 is the dividend.

domain

The set of allowable values of x in a function or relation.

Extended in Mathematics Advanced, Extension 1 and Extension 2: For a function or relation, it is the set of real numbers on which the function or relation is defined.

doubling and halving

Doubling and halving is a strategy that can make solving multiplication problems easier, e.g. solving 4 x 12 is the same as solving 2 x 24.

Doubling and halving is also a strategy used to find factors of a number. For example, to find the factors of 24, start with 1 x 24 then double and halve, 2 x 12, 4 x 6, 8 x 3.

edge

The line segment formed by the intersection of two faces of a polyhedron, e.g. a cube has 6 faces and 12 edges.

edge (networks)

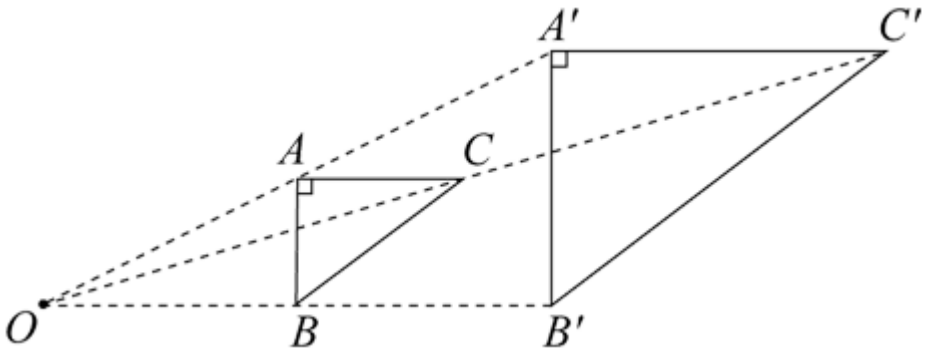
A line that joins vertices to each other in a network diagram.

element(s)

A member of a set. For example, *3* is a member of the set of natural numbers $N = \{0,1,2,3,4,\ldots\}$. This relation can be written more concisely as $3 \in N$ ('*3* is an element of the set N').

enlargement

A type of transformation that increases or decreases the size of an image without changing its shape, depending on the size of the enlargement factor.



equal-arm balance

A balance where the distance from the support point (middle) to the arms at each end is equal.

equally likely outcomes

An event for which all outcomes have the same probability of occurring. For example, in tossing a fair coin, the outcome 'head' and the outcome 'tail' are equally likely. In this situation, P(head)=P(tail)=0.5.

equals sign

In mathematics, the equals sign is used to indicate equivalence, e.g. 5 + 1 = 2 + 4 and to show that two or more amounts have the same value.

See equivalent

equation

An expression showing the equality of 2 quantities, using the = sign between them. A mathematical formula asking for a solution so that the 2 expressions in that variable are equal, e.g. $x^2 - 1 = x$.

equiangular

When the angles of one polygon are equal to the corresponding angles of another polygon.

equivalent

Two things are equivalent if they have the same value.

See equal, equals sign.

Extended in Mathematics Advanced, Extension 1 and Extension 2: Two things are equivalent if they have the same value, function or meaning. For example, $y = a^x$ and $x = \log_a y$ are equivalent. Statements P and Q are equivalent if P implies Q and Q implies P.

equivalent fractions

Fractions that are different representations of the same value, e.g. $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$.

Two fractions $\frac{a}{b}$ and $\frac{c}{d}$ are equivalent if they are equal, that is, $ad = bc$.

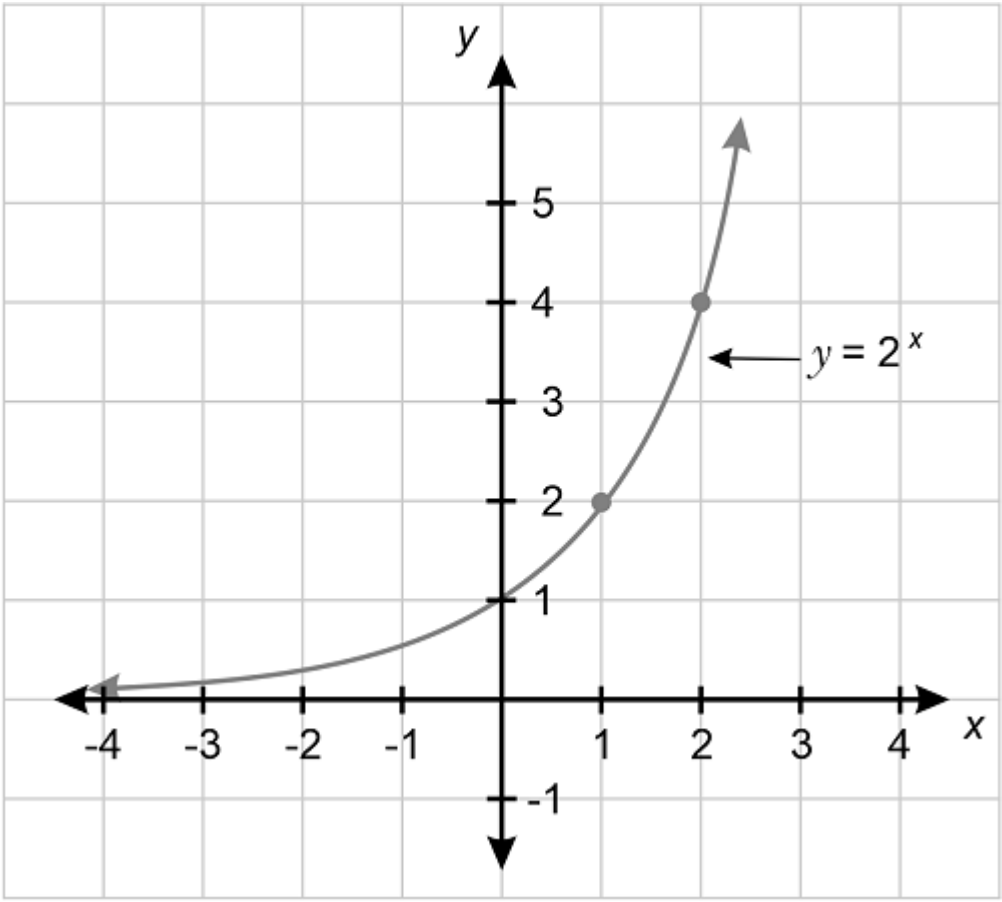
event

Situations that occur in the everyday context, e.g. having lunch. In the context of probability, the set of possible outcomes.

Extended in 7–10: A subset of the sample space for a random experiment. For example, the set of possible outcomes from tossing 2 coins is $\{HH, HT, TH, TT\}$, where H represents a 'head' and T is a 'tail'. For example, if A is the event 'at least one head is obtained', then $A = \{HH, HT, TH\}$.

exponential relationship

A relationship in which the independent variable occurs as an exponent (or power/index) with a positive base. For example, $y = 2^x$ is an exponential relationship where x is the independent variable. The graph of an exponential relationship is an exponential curve.



expression

Two or more numbers or variables connected by operations. For example, $17 - 9$, $8 \times (2 + 3)$, $2a + 3b$ are all expressions. Expressions do not include an equals sign.

Extended in Mathematics Advanced, Extension 1 and Extension 2: The result of combining numbers, variables and operations in a meaningful way. For example, $\frac{x^3-1}{\sin y}$ is an expression, but $(x^3-)\int / \sin y$ is not a validly formed expression.

extrapolate/extrapolation

Occurs when the fitted model, such as a line of best fit, is used to make predictions using values that are outside the range of the original data upon which the fitted model was based.

face

A flat surface of a 3-dimensional geometric object (polyhedron) with only straight edges.

face (networks)

A bounded region in a network.

factor

A number that divides another number without a remainder. For example, 1, 2, 3 and 6 are factors of 6 but 4 and 5 are not.

Extended in 7–10: Numbers or algebraic expressions are factors (or divisors) of another number if they multiply to give that number. For example, 3 and 4 are factors of 12 as $3 \times 4 = 12$. This can be written algebraically as x and y are factors of m, if $m = xy$.

factor theorem

If $p(x)$ is a polynomial and $p(a) = 0$ for some number a, then $p(x)$ is divisible by $x - a$. The factor theorem can be used to obtain factors of a polynomial.

factorise

To express a number or algebraic expression as a product. For example, 15 is factorised when expressed as a product: $15 = 3 \times 5$ and $x^2 - 3x + 2$ is factorised when written as a product: $x^2 - 3x + 2 = (x + 1)(x - 2)$.

five (5)-number summary

A method for summarising a dataset using 5 statistics: the minimum value, the lower quartile, the median, the upper quartile and the maximum value.

fraction

The result of a division. For example, 2 divided by 4 produces quarters. Fractions also represent:

- a relationship or ratio
- parts when a whole is partitioned into equal parts
- numbers between whole numbers
- units of measure.

frequency

The number of times that a particular value occurs in a dataset. For grouped data, it is the number of observations that lie in that group or class interval. For example, when rolling a dice 20 times, ‘the frequency of a 6’ means how many times the number 6 comes up.

frequency distribution

A visual display that organises and presents frequency counts of scores so that the information can be interpreted more easily.

Frequency distributions can be displayed in tabular or graphical form.

function

A function f assigns to each element of one set S precisely one element of a second set T.

The functions most commonly encountered in elementary mathematics are real functions of real variables. For such functions, the domain are sets of real numbers.

Functions are usually defined by a formula for $f(x)$ in terms of x. For example, the formula $f(x) = x^2$, defines the 'squaring function' that maps each real number x to its square x^2.

future value

The total value of an investment or annuity at the end of a specified term, including all contributions and interest earned.

gradient

The slope of a line. It is calculated as the gradient of a line segment it contains. If $A(x_1, y_1)$ and $B(x_2, y_2)$ are 2 distinct points on a line, the gradient of the line (or line segment AB) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

graph

A visual representation of statistical data or of a relationship between variables. Ordered pairs of values (x, y) that represent the function or relation are plotted to form a graph. Graphs of statistical data include dot plots, box plots, column graphs, divided bar graphs and histograms.

Extended in Mathematics Advanced, Extension 1, Extension 2: For a function f, its graph consists of all ordered pairs $(x, f(x))$ for x in the domain of f.

Gregorian calendar

The Gregorian calendar is the most widely used solar dating calendar. It was introduced in 1582 as a modification of the Julian calendar.

grouping division

Grouping (quotitive) division requires finding how many groups are formed, e.g. if I have 12 marbles and each child gets 4, how many children will get marbles?

When grouping, the quotient represents the number of groups within the shared quantity.

hectare

A common unit of land measure in the metric system equal to 10 000 square metres (approximately 2.47 acres).

hefting

Testing the weight of an object by lifting and/or balancing it.

hemisphere

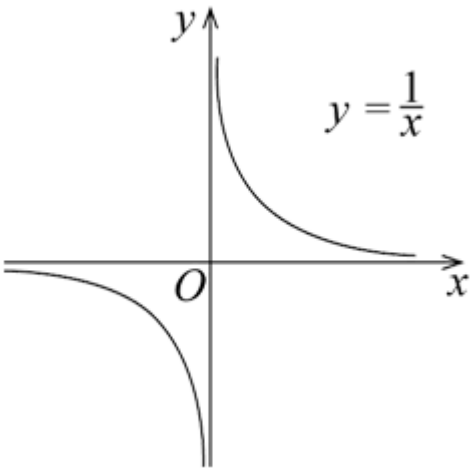
Half of a sphere.

highest common factor (HCF)

The common divisor of a given set of natural numbers that is greater than each of the other common divisors. For example, 1, 2, 3 and 6 are common factors of 24, 54, and 66, and 6 is the highest common factor.

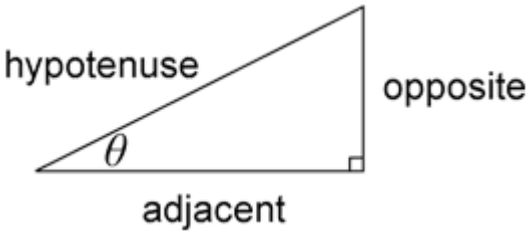
hyperbola

The graph of $y = \frac{1}{x}$ produces what is called a rectangular hyperbola. The x- and y- axes are asymptotes.



hypotenuse

The longest side in a right-angled triangle that is opposite the right angle.



hypothesis

A supposition or proposed explanation made on the basis of limited evidence as a starting point for further investigation.

independent events

Two events are independent if knowing the outcome of one event tells us nothing about the outcome of the other event.

independent variable

A variable used to represent values in the domain (input values) of a function. Generally represented on the horizontal axis of a graph.

index

Also called ‘exponent’. The power to which a number or algebraic expression is to be raised. The index or exponent is written as a superscript. Positive integral exponents indicate the number of times a term is to be multiplied by itself.

indices

The plural of the term index.

inequality

A statement that one number or algebraic expression is less than (or greater than) another. There are 4 types of inequalities:

- a is less than b is written $a < b$
- a is greater than b is written $a > b$
- a is less than or equal to b is written $a \leq b$
- a is greater than or equal to b is written $a \geq b$.

Extended in Advanced, Extension 1, Extension 2: An order relation between one number or algebraic expression and another.

inference

A conclusion reached on the basis of evidence and reasoning.

infographic

A chart, diagram or illustration (as in a book or magazine or on a website) that uses graphic elements to present information in a visually striking way.

integer

A whole number, positive, negative or zero, e.g. −3, −2, −1, 0, 1, 2 …

Extended in Advanced, Extension 1, Extension 2: The set of integers is usually denoted by $\mathbb{Z}$.

intercept

The point at which a curve or function crosses an axis or other curve in a plane. The point at which a curve crosses the x-axis ($y = 0$) is called the x-intercept and the point at which a curve crosses the y-axis ($x = 0$) is called the y-intercept.

Extended in Advanced and Extension 1: The x- and y-intercepts are sometimes taken to mean the signed distance from the point at which the curve crosses the axis to the origin, e.g. for the line $y = mx + c$, the y-intercept is c.

interior angle

When referring to polygons, the inner angles formed by 2 adjacent sides.

When referring to a transversal cutting through a pair of straight lines, the 4 angles formed are called interior angles.

internal volume (capacity)

A measure of how much a container can hold.

interpolate/interpolation

Making predictions between known data values. For example, working between 2 known points on a graph to predict a value in between these points.

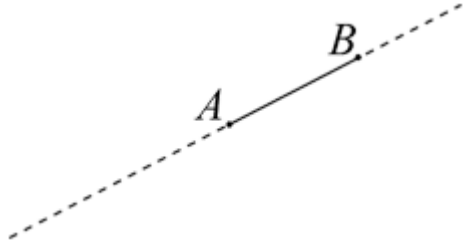
interquartile range (IQR)

A measure of the spread within a numerical dataset. It is equal to the upper quartile (Q_3) minus the lower quartile (Q_1); that is, $IQR = Q_3 - Q_1$.

The IQR is the width of an interval that contains the middle 50% (approximately) of the data values.

interval

A continuous subset of the real number line. For example, ‘the set of all real numbers greater than or equal to 10’.



inverse operation

The operation that reverses the effect of another operation.

Examples:

- Addition and subtraction are inverse operations. When you add 3 to 7 you get 10. If you then subtract 3, you get back to 7.
- Multiplication and division are inverse operations. When you multiply 6 by 2 you get 12. If you then divide by 2 you get back to 6.

inverse variation

When one variable increases as the other variable decreases. For example, if y is said to be ‘inversely proportional’ to x, the equation is of the form $y = \frac{k}{x}$, where k is a constant of variation (or proportion). Also known as inverse proportion.

irrational number

A number that cannot be expressed as a fraction in the form $\frac{m}{n}$, where m and n are integers and n is non-zero. The decimal form of irrational numbers does not terminate and is non-recurring.

jump strategy

A mental calculation method that involves jumping from one number (usually the largest number) either forwards (addition) or backwards (subtraction) to the solution.

length

Measure of an object from end to end.

line

A one-dimensional figure that is straight, has no thickness and extends endlessly.

line of best fit

A line drawn through a scatter plot of data points that represents the nature of the relationship between 2 variables.

linear equation

An equation involving linear expressions. The general form of a linear equation in one variable is $ax + b = c$ where a, b and c are constants.

linear function

y is a linear function of x if $y = mx + c$ where m and c are constants.

The graph of a linear function is a straight line, where m is the gradient and c is the y-intercept.

See linear relationship

linear pattern

A pattern created by the regular repetition of units with the same difference between terms.

linear relationship

Two variables x and y are in a linear relationship (or form a linear function) if they are connected by an equation of the form $y = mx + c$. Graphically, m is the gradient and c is the intercept with the vertical axis of the corresponding linear graph.

list

A number of items, typically presented one after the other.

literal equation

An equation that contains mostly letters, such as $P = 2l + 2w$, where each variable represents a quantity 'literally'. Formulas are literal equations.

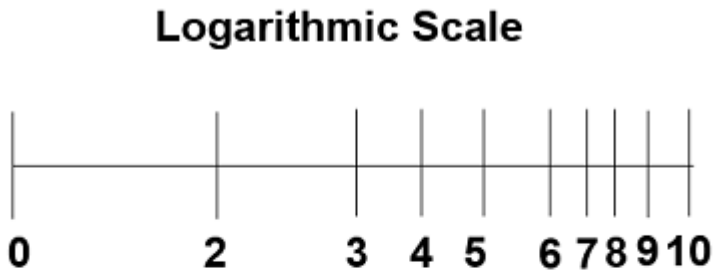
logarithm

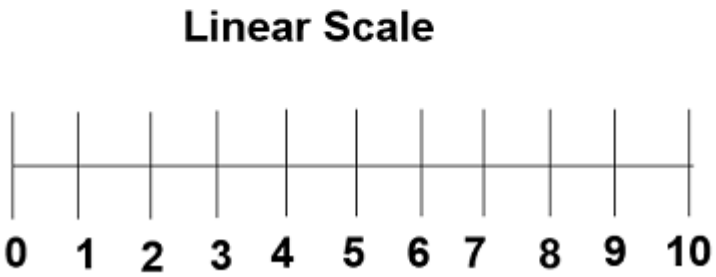
The logarithm of a positive number x is the power to which a given number b, called the base, must be raised in order to produce the number x. The logarithm of x, to the base b is denoted by $\log_b x$.

Algebraically: $\log_b x = y$ is equivalent to $b^y = x$.

logarithmic scale

A scale where successive endpoint values of intervals increase by a constant factor (multiplicatively). Contrast with linear scales in which the increase is a constant amount.





magnitude

The size or absolute value of a number. For example, $+4$ and -4 have a magnitude of 4.

Extended in Mathematics Extension 1 and Mathematics Extension 2: For a vector, the magnitude is its length, given by $|x\mathbf{i} + y\mathbf{j}| = \sqrt{x^2 + y^2}$ for 2D vectors and $|x\mathbf{i} + y\mathbf{j} + z\mathbf{k}| = \sqrt{x^2 + y^2 + z^2}$ for 3D vectors.

mass

The amount of matter in an object.

mean

The sum of values in a dataset divided by the total number of values in the dataset. Also called the average.

measure of centre

A statistic that is used to summarise a dataset. There are 3 common measures of centre for a dataset: mode, median and mean.

measure of spread

In statistics, different methods of calculating the variability of a set. The most commonly used measures of spread are the range, interquartile range and standard deviation.

median

The value in a set of ordered data that divides the data into 2 parts. It is frequently called the 'middle value'.

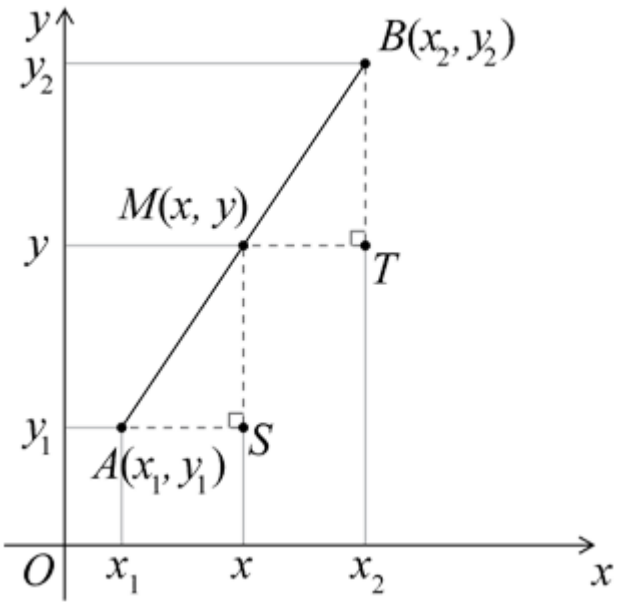
Extended in Mathematics Standard and Advanced: If the number of values in the ordered dataset is even, it is the average of the two middle values. If the number of values in the dataset is odd, the median is the middle value.

midpoint

A point on a line segment or interval that divides the segment into 2 equal parts.

Let (x_1, y_1) and (x_2, y_2) be points in the Cartesian plane. Then the midpoint M of line segment AB has coordinates $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

This can be seen from the congruent triangles below.



mode

The most frequently occurring value in a set of data.

Extended in 7–10: There can be more than one mode. When there are 2 modes, the dataset is said to be bimodal.

monic

A polynomial in which the coefficient of the leading term is 1.

multiples

Products formed using the same base number multiplied by different whole numbers, e.g. 3, 6, 9, 12 …

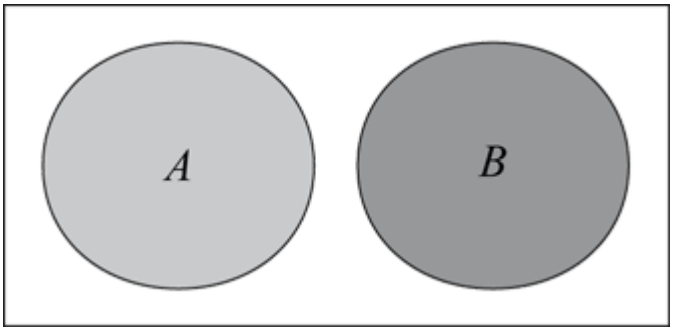
multiplicity (of a root/zero)

The number of times a particular number is a zero for a given polynomial or a root for a polynomial equation.

mutually exclusive events

Two events that cannot have simultaneous outcomes in the same chance experiment. For example, when a fair coin is tossed twice, the events '*HH*' and '*TT*' cannot occur at the same time and are, therefore, mutually exclusive.

In a Venn diagram, as shown below, mutually exclusive events do not overlap.

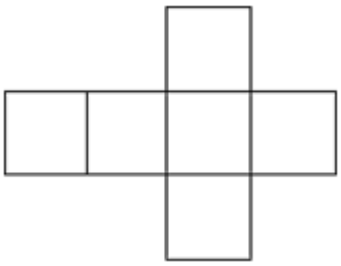


Extended in Mathematics Advanced: For a random experiment two events A and B are mutually exclusive if $A \cap B = \emptyset$, that is, they cannot occur simultaneously.

net

A diagram showing the pattern of connected faces and the edges of an unfolded polyhedron (a solid with all faces polygons).

One possible net for a cube is shown below.



network

A set of points (vertices or nodes) some of which are joined by lines or curves (edges) that sometimes enclose regions (faces). For example, road networks, a family tree or the edges lining a tennis court.

non-linear

Functions or graphs that cannot be represented by a straight line or a linear function.

number bonds

Combinations (pairs) of numbers that add to a given number, e.g. 8 + 2, 6 + 4, and 7 + 3 all bond to form 10.

number chart

A table that lists numbers in numerical order. There are different types of number charts.

For example:

- a 1–120 or 0–99 chart
- a calendar display for a month.

number line

A number line is used to represent numbers according to their distance from a zero point. The representation of a number line can start and end on any number.

number sentence

A number sentence uses numerals and mathematical symbols. A number sentence may be used instead of the word equation. For example, instead of writing 6 apples plus 7 apples equals 13 apples, the number sentence would be 6 + 7 = 13.

numerator

In the fraction $\frac{a}{b}$, a is the numerator. If an object is divided into b equal parts, then the fraction $\frac{a}{b}$ represents a of these parts all together. For example, if a line segment is divided into 5 equal parts, each of these parts is one-fifth ($\frac{1}{5}$) of the whole and 3 of these parts taken together corresponds to the fraction $\frac{3}{5}$.

numerical variable

Variables whose values are numbers and for which arithmetic processes, such as adding and subtracting or calculating an average, make sense.

A discrete numerical variable is a numerical variable, each of whose possible values is separated from the next by a definite 'gap'. The most common numerical variables have the counting numbers 0, 1, 2, 3, … as possible values. Others are prices, measured in dollars and cents. For example, the number of children in a family or the number of days in a month.

object

A three-dimensional solid.

obtuse angle

An angle bigger than a right angle (90°) but smaller than a straight angle (2 right angles, or 180°).

one-to-one data display

Display of data using objects and pictures where one object or picture represents one data value.

operation

The process of combining numbers or expressions. Operations are arithmetic – addition, subtraction, multiplication and division – and also include exponentiation and substitution.

origin

The point of intersection of the x- and y-axes on the Cartesian plane.

outcome

Possible result from an experiment or trial.

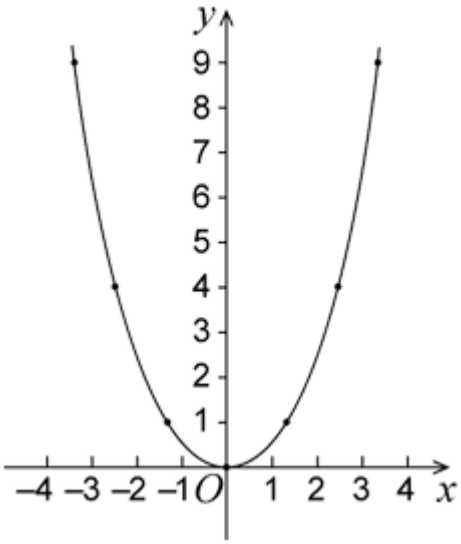
outlier

A data value that appears to stand out from the other members of the dataset by being unusually high or low.

parabola

The graph of $y = x^2$. The point $(0,0)$ is called the vertex of the parabola and the y-axis is the axis of symmetry of the parabola.

Some other parabolas are the graphs of $y = ax^2 + bx + c$ where $a \neq 0$.



parallel

Two distinct lines, rays or line segments in the same plane that have no points of intersection and so necessarily have the same gradient (slope).

The symbol ∥ is used to express that one line, ray, or line segment is parallel to another.

partition

Partitioning is dividing a quantity into parts.

In the early years, it commonly refers to treating whole numbers as made up of 2 parts (number bonds), e.g. 10 is 8 and 2.

In later years, it refers to dividing both continuous and discrete quantities into equal parts (equi-partitioning) when learning about fraction and division concepts.

path (networks)

A sequence of consecutive edges in a graph. The length of the path is the number of edges traversed.

pattern

A pattern is made up of a number of elements that repeat or follow a rule.

pentagon

A polygon with 5 sides.

See polygon

percentage error

The absolute error of a measurement expressed as a percentage of the recorded measurement.

perpendicular

Two lines, rays, line segments, vectors, planes or other objects that intersect at a 90° angle (a right angle).

pi

The ratio of the circumference of any circle to its diameter. It is denoted by the Greek letter π (pi). Approximate values are *3.14* or $\frac{22}{7}$.

picture graph

Display of data using images, symbols or pictures to represent data in categories.

piecework

Employment where a worker is paid a fixed rate for each item produced or action performed regardless of the time taken.

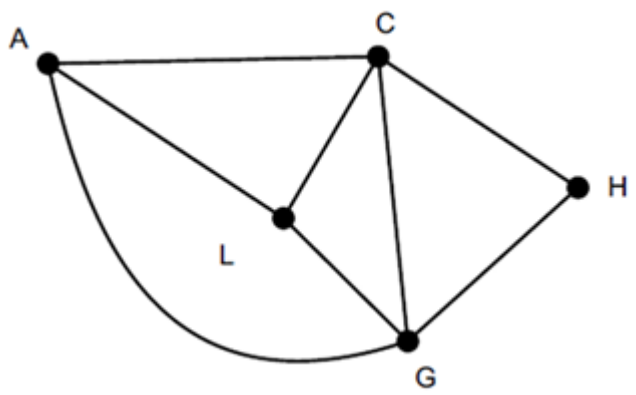
place value system

The place value system has 4 properties: positional, base-ten, additive and multiplicative.

The value of a digit is determined by its position in a number relative to the ones (or units) place. For example, in the number 924, the 4 denotes 4 ones, the 2 denotes 2 tens or 20 ones, and the 9 denotes 9 hundreds, 90 tens or 900 ones.

planar graph

A connected graph that is drawn within a plane whose edges do not cross.



point

Representation of a location in space that has zero dimensions. Lines, curves, shapes, surfaces and objects are constructed from sets of points. Points in the Cartesian plane are specified by coordinates.

polygon

Plane shape bounded by 3 or more line segments.

polynomial

An expression made up of non-negative integer powers of the same variable and coefficients combined using addition, subtraction and multiplication.

population

The complete set of individuals, objects, places, etc., that we want information about.

Extended in Mathematics Standard and Mathematics Extension 1: In statistics it is the entire dataset from which a statistical sample may be drawn.

prime factor

A prime factor of a natural number n is a factor of n that is a prime number. For example, the prime factors of 330 are 2, 3, 5 and 11.

principal

The total amount of money borrowed or invested on which interest is calculated. Also referred to as present value (PV).

probability

The chance of something happening shown on a scale from 0 and 1 (inclusive). For example, the probability that a fair coin toss will come up 'heads' is 0.5.

pronumeral

A letter or symbol that is used to represent a value in a problem that can vary or change. Also known as a variable.

proof

A rigorous mathematical argument that demonstrates the truth of a given statement or proposition. A mathematical statement that has been established by means of a proof is called a theorem.

proportion

A constant ratio between corresponding elements of 2 sets. For example, the circumference and diameter of a circle are in proportion as their ratio is the constant π (pi).

Pythagoras' theorem

The square of the length of the hypotenuse, c, of a right-angled triangle equals the sum of the squares of the lengths of the other 2 sides, a and b, such that $c^2 = a^2 + b^2$.

quadratic equation

An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$, b and c are constants.

quadratic expression

An expression of the form $ax^2 + bx + c$, where $a \neq 0$, b and c are constants.

quadratic formula

The roots of a quadratic equation $ax^2 + bx + c = 0$ where $a \neq 0$ are given by the quadratic formula:

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

quantity value

The amount a digit represents in a number. For example, the quantity value of the five in 546 is 500, and the quantity value of the four is 40.

quartiles

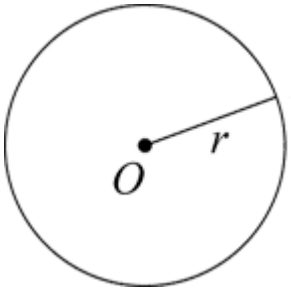
The values that divide an ordered dataset into 4 (approximately) equal parts. There are 3 quartiles. The first, the lower quartile (Q_1), divides off (approximately) the lower 25% of data values. The second quartile (Q_2) is the median. The third quartile, the upper quartile (Q_3), divides off (approximately) the upper 25% of data values.

quotient

The result of dividing one number or algebraic expression by another.

radius

The distance from the centre of a circle to any point on its perimeter. It is equal to half of the circle's diameter.



random number

A number whose value is governed by chance. For example, the number of dots showing when a fair die is tossed. The value of a random number cannot be predicted in advance.

range (function)

The set of values of the dependent variable for which a function is defined.

range (of a dataset)

The difference between the highest and lowest values in a dataset.

ratio

A comparison of magnitudes of sets, quantities of the same kind, or algebraic expressions. It is often used as a comparison $a : b$ of the size of 2 (or more) quantities relative to each other.

rational number

An element of the infinite set of numbers $Q = \{\frac{m}{n}$, where m and n are integers and $n \neq 0\}$. It may be expressed in decimal form, e.g. $\frac{1}{8} = 0.125$ and $\frac{4}{9} = 0.444$ ...

ray

The part of a line that starts at a point and continues in a particular direction to infinity.

reciprocal

If x is any real number, then the reciprocal of that number will be $\frac{1}{x}$. For example, the reciprocal of 4 is $\frac{1}{4}$.

rectangle

A four-sided shape (quadrilateral) with opposite sides of equal length and all 4 interior angles equal (right angles).

recurring decimal

A decimal expansion of a number that contains a finite sequence of digits that repeats indefinitely, such as *34.271271271* ... The repeating sequence is often designated by a bar over it, such as $34.\overline{271}$. Every recurring decimal is the decimal expansion of a rational number.

reflection

A transformation of a shape formed by creating a mirror image on the other side of a given line.

reflex angle

An angle that measures between 180 and 360 degrees, i.e. between 2 and 4 right angles.

regular shape

A two-dimensional polygon with all sides and all interior angles equal. A square is an example of a regular shape.

relation

An association between the elements of one set, A, and the elements of another set, B. It may be represented as a set of ordered pairs (a, b), where a is in A, b is in B and b is related to a.

relative frequency

Given by the ratio $\frac{f}{n}$, where f is the frequency of occurrence of a particular data value or group of data values in a dataset, and n is the number of data values in the dataset.

remainder

The part 'left over' when dividing a number into equal groups. The remainder forms part of the next group.

Extended in 7–10: The amount left over when one number or algebraic quantity a is divided by another b. If a is divisible by b, then the remainder is 0. For example, when 68 is divided by 11, the remainder is 2, because 68 can be expressed as $68 = 6 \times 11 + 2$.

Extended in Mathematics Extension 1: What is left over after a division. When division of a polynomial $P(x)$ by a non-zero polynomial Q(x) gives $P(x) = A(x)Q(x) + R(x)$, where the degree of $R(x)$ is less than the degree of $Q(x)$, then $R(x)$ is the remainder.

repeated addition

An early strategy for solving multiplication problems used to add groups of the same quantity. For example, 3 + 3 + 3 + 3 to solve 4 groups of 3.

rhythmic counting

Rhythmic counting refers to the repeated sound pattern that students may use to vocalise the count – creating a beat to emphasise the pattern.

rounding

The process for approximating a value that lies between 2 known values. It is used to specify a number correct to a given accuracy. For example, 4.027 becomes 4.03 when rounded to 2 decimal places.

sample

A subset of a population used to estimate characteristics of the population. For example, a randomly selected group of 8-year-olds (sample) selected to estimate the height of 8-year-olds in Australia (population).

sample space

The set of all possible outcomes of a chance experiment. For example, the set of outcomes (also called sample points) from tossing 2 identical coins at the same time is {HH, HT, TT}, where H represents a ‘head’ and T a ‘tail’.

scatter plot

A visual representation of bivariate numerical data using the position of each dot on the horizontal and vertical axis to indicate the numerical values for an individual data point. Scatter plots are used to observe relationships between variables.

scientific notation

A way of writing numbers that are too big or too small to be written in an accessible way. It uses powers of 10. For example, the scientific notation for 34 590 is 3.459×10^4.

secant

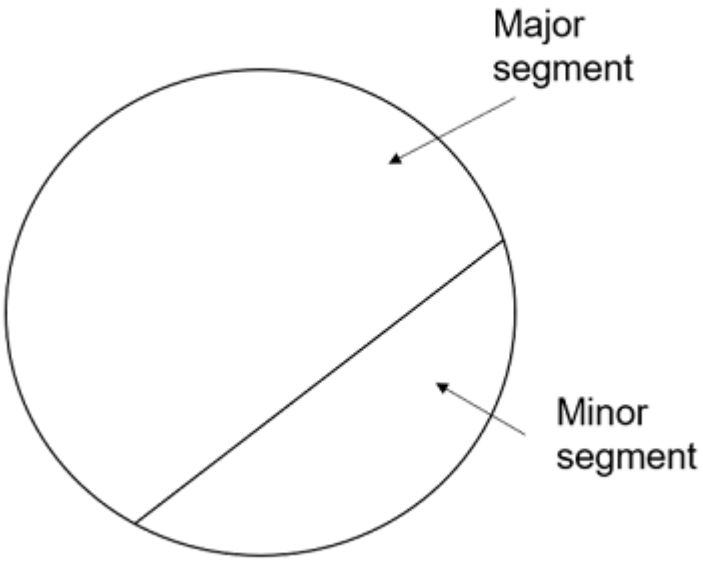
The straight line passing through 2 points on the graph of a function.

sector

The plane figure enclosed by 2 radii or a circle and the arc between them.

segment

A plane figure enclosed by a chord and the arc joining the endpoints of a chord.



set

A collection of objects or elements, usually specified by listing its elements, e.g. $\{1,2,3,4\}$; by describing it in words, e.g. 'the set of primes'; or by using a rule such as $\{(x, y): y = 2x + 1, x \in N\}$.

Extended in Mathematics Advanced, Extension 1 and Extension 2: A collection of distinct, unordered objects, referred to as members or elements of the set.

shape

A two-dimensional closed figure.

sharing division

Sharing (partitive) division requires finding how many are in each group, eg if 12 marbles are shared between 3 students, how many does each student get?

When sharing, the quotient represents the quantity of shared objects in each group.

side

A side is the line segment joining 2 vertices of a two-dimensional shape.

significant figures

Each of the digits of a number that are used to express it to the required degree of accuracy, starting from the first non-zero digit. For example, $0.000034 = 3.4 \times 10^{-5}$ has 2 significant figures and when 123,456 is rounded to 3 significant figures, the result will be 123 000.

similarity

When the enlargement of one figure is congruent to another figure. Similar figures have corresponding lengths in the same proportion, are the same shape and have equal corresponding angles.

simple interest

The interest accumulated when the interest payment in each period is a fixed percentage of the principal (the initial lump sum of money).

The simple interest formula is given by $I = PRn$ where I is the interest earned, P is the principal value invested, R is the rate of interest and n is the number of time periods over which the interest is applied.

simulation

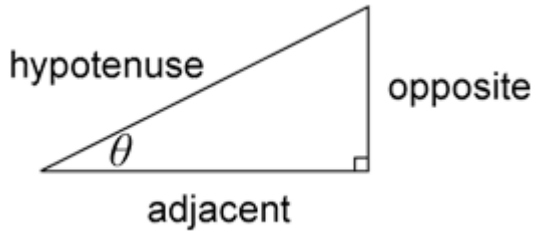
A recreation of random phenomena used to collect probability data through either physical or virtual manipulatives such as spinners, coins and cards or computer-generated simulations that can be performed a large number of times.

simultaneous equations

A set of 2 or more equations, each containing 2 or more variables whose values can simultaneously satisfy all the equations in the set, the number of variables being equal to or less than the number of equations in the set.

sine ratio

In any right-angled triangle, $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$ where $0^\circ < \theta < 90^\circ$.



skewness

A measure of asymmetry (non-symmetry) in a distribution of values about the mean of a set of data.

skip counting

Skip counting is counting forwards or backwards in groups or multiples of a particular number, e.g. 5, 10, 15, 20 ...

slide (translation)

Sliding a shape without turning it.

square

A four-sided shape (quadrilateral and a parallelogram) with all sides of equal length and all 4 interior angles equal (right angles).

square numbers

The result of multiplying a natural number by itself. For example, $16 = 4 \times 4 = 4^2$, and is an element of the set $\{0, 1, 4, 9, 16, 25\ldots\}$.

square root

A value that when multiplied by itself gives the original number. The positive square root of a given real number x is the positive real number y such that $y^2 = x$.

standard deviation

A measure of the variability or spread of a dataset. It gives an indication of the degree to which the individual data values are spread around their mean.

Extended in Mathematics Standard and Advanced: For a random variable, the standard deviation is the square root of its variance.

It provides a measure of the spread of the probability density function.

stem-and-leaf plot

A method of organising and displaying numerical data in which each data value is split into 2 parts, a 'stem' and a 'leaf'. For example, the stem-and-leaf plot below displays the resting pulse rates of 19 students. The stem is on the left side of the vertical line and the leaves are on the right side.

pulse rate

6	8 8 8 9
7	0 1 1 4 6 6 8
8	2 6 8 8
9	0 6
10	4
11	0

See back-to-back stem-and-leaf plot

subtend

In geometry, an angle subtended by an arc or interval is the angle whose 2 rays pass through the endpoints of the arc or interval. A possible synonym for ‘subtends’ is ‘makes’.

superimposed

To place or lay on top of.

supplementary angles

Two adjacent angles that form a straight angle and whose sum is 180 degrees (a straight angle). For example, 45° and 135° are supplementary angles.

surd

A numerical expression involving one or more irrational roots of numbers.

Extended in Mathematics Advanced: An irrational root of a number, e.g. $\sqrt{5}$ and $\sqrt[3]{4}$.

More generally sums involving rational multiples of surds are sometimes also referred to as surds, e.g. $2 + \frac{3\sqrt{7}}{4}$.

surface

The flat or curved area of an object. A surface with straight edges is called a face, a surface with a curved boundary is called a curved surface.

See face

surface area

The measure of the total area of the surface(s) of a three-dimensional shape or object. For example, the surface area of a cube with side length of 5 units is 150 square units.

symmetrical distribution

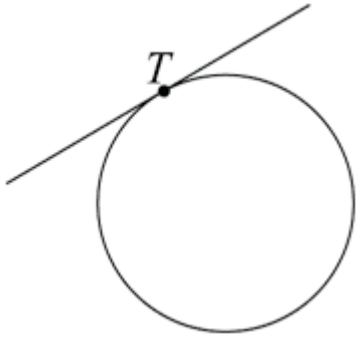
When the 2 sides of the distribution are a mirror image of each other. A normal distribution is a true symmetric distribution of observed values.

table

An arrangement of information in rows and columns.

tangent

A line that intersects a circle at just one point. It touches the circle at that point of contact but does not pass inside it.

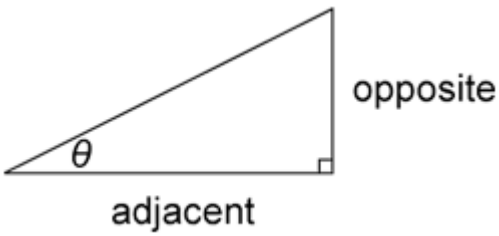


Extended in Mathematics Advanced, Extension 1 and Extension 2: For a curve at a given point P the tangent can be described intuitively as the straight line that 'just touches' the curve at that point. At P the curve has 'the same direction' as the tangent. In this sense, it is the best straight-line approximation to the curve at point P. For a differentiable function $f(x)$ the tangent to its graph at the point $P(a, f(a))$ has slope $f'(a)$.

tangent ratio

The ratio of the length of the altitude, a, of a right-angled triangle to the length of its base, b, for a given base angle, theta, θ.

In any right-angled triangle, $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$, where $0° < \theta < 90°$.



taxable income

The amount of yearly income that is used to calculate an individual's or company's payable income tax.

terminating decimal

A decimal expansion of a number that has a finite number of non-zero digits, e.g. 3.125. Every terminating decimal represents a rational number where the denominator is a power of 10.

trail (networks)

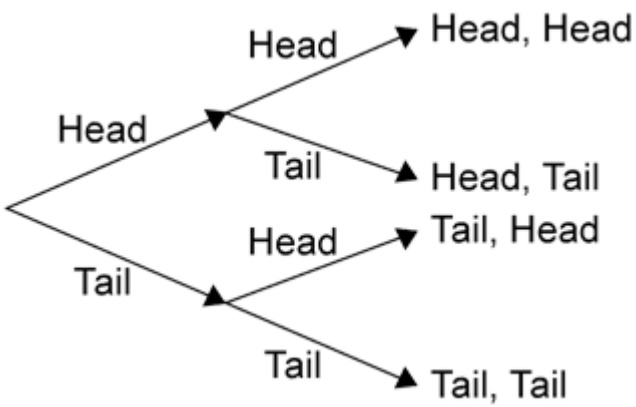
A walk where no edges are repeated.

transversal

A line that intersects 2 other lines obliquely.

tree diagram

A diagram consisting of line segments (edges) connected to points (vertices) like the branches of a tree. It shows the relationship between sets, events or the set of outcomes of a multi-step random experiment.



See probability tree

triangle

A closed shape with exactly 3 straight sides and 3 interior angles.

trigonometric ratios

The relationship between the angles and sides of right-angled triangles. The 3 basic trigonometric ratios are sine, cosine and tangent.

See sine ratio, cosine ratio and tangent ratio

true bearing

Measured in degrees clockwise from true north and written with 3 digits to specify the direction. For example, the direction of north is specified as *000*°T, east is specified as *090*°T, south is specified as *180*°T and north-west is specified as *315*°T.

two-dimensional

Having length and width only (with no depth).

two-way table

A common way of displaying the two-way frequency distribution that arises when a group is categorised according to 2 criteria.

	Curly hair	Straight hair	Total
Red hair	1	1	2
Brown hair	8	4	12
Blonde hair	1	3	4
Black hair	7	2	9
Total	17	10	27

uniform

Means ‘the same as’. Used to refer to units of the same size used for measuring.

unit circle

A circle with a radius of 1 unit.

variable (algebra)

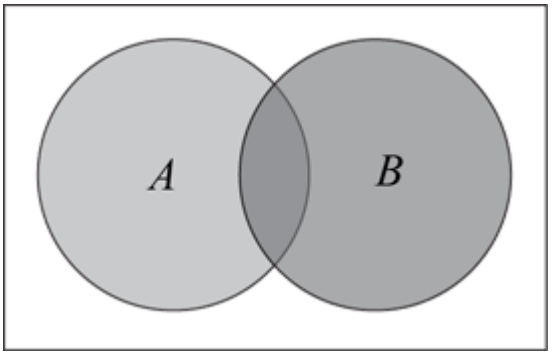
Things that are measurable or observable that are expected to either change over time or between individual observations. They are often designated by symbols, such as n, x and y, to represent members of a set. For example, the variable x could represent an unspecified real number.

variable (statistics)

Something measurable or observable that is expected to change either over time or between individual observations. For example, the age of students, their hair colour or a playing field's length or its shape.

Venn diagram

Graphical representations, using several typically overlapping circles, showing elements of sets in relation to properties or attributes. They are drawn for some specified universal set.



vertex

Where 2 straight sides of a two-dimensional shape meet.

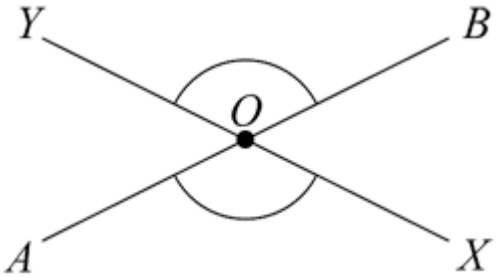
Extended in 7–12: A vertex is a point in the plane where lines meet and do not extend beyond, or a point in space where several edges meet. A vertex can also refer to a node in a network. It is also the turning point of a parabola.

vertical line test

Determines whether a relation or graph is also a function. If a vertical line intersects or touches a graph at more than one point, then the graph is not a function.

vertically opposite angles

Two pairs of vertically opposite angles formed at the point of intersection of 2 lines in a plane. Vertically opposite angles are equal.



volume

The amount of space occupied by an object.

walk (networks)

A sequence of edges and vertices in a network.

weight

The measure of the force of gravity acting on a body with a given mass. An object can be weightless without its mass being zero. The term ‘weight’ is used as a synonym for mass when the concept of gravitational force has not been introduced.

Global Terms

Aboriginal and Torres Strait Islander Peoples

Aboriginal Peoples are the first peoples of Australia and are represented by more than 250 language groups, each associated with a particular Country or territory. Torres Strait Islander Peoples are represented by 5 major island groups, and are associated with island territories to the north of Australia's Cape York which were annexed by Queensland in 1879.

An Aboriginal and/or Torres Strait Islander person is someone who:

- is of Aboriginal and/or Torres Strait Islander descent
- identifies as an Aboriginal person and/or Torres Strait Islander person, and
- is accepted as such by the Aboriginal and/or Torres Strait Islander community(ies) in which they live.

Aboriginal English

A recognised dialect of English which is the first, or home language, of many Aboriginal people. It differs from other dialects of English, such as Standard Australian English, in systematic ways including sounds, grammar, words and their meanings, and language use. Aboriginal English is a powerful vehicle for the expression of Aboriginal identity. Aboriginal English is not a target language study option for NSW Aboriginal Languages syllabuses.

Aboriginal Songline(s)

Texts that describe landscapes and directions of the tracks forged in lands, waters and skies by Creator Spirits during the Dreaming.

accessibility

The extent to which a system, environment or object may be used irrespective of a user's capabilities or abilities. For example, the use of assistive technologies (AT) to allow people with disability to use computer systems, or the use of icons in place of words to allow young children to use a system.

assistive technology (AT)

A device or system whose primary purpose is to maintain or improve an individual's functioning and independence to facilitate participation and enhance overall wellbeing. This includes technologies specifically designed to meet an individual's needs, eg eye gaze technology, as well as more general technologies that can be used by anyone, eg speech-to-text applications. Assistive technology can also be referred to as inclusive technology.

augmentative and alternative communication (AAC)

An umbrella term that encompasses the communication methods used to supplement or replace speech or writing. AAC can be unaided, such as gestures, body language and sign language, or aided such as pictures, symbols, objects or speech generating devices.

See communication systems

communication forms

The ways people communicate and the communicative behaviours they use. Communication forms can be non-symbolic and/or symbolic. Non-symbolic forms include sounds, gestures, facial expressions and eye movements. Symbolic forms can be aided or non-aided. Aided forms of symbolic communication include objects, symbols, photographs and drawings. Aided forms can be

digital. Non-aided forms of symbolic communication include formal gestures; speech; and signs, such as Key Word Sign.

copyright

The protection provided to the creators of original works and makers of sound recordings and films, that offers a legal framework for the control and reproduction or transmission of their literary, dramatic, artistic or musical works.

Country/Place

Country is used to describe a specific area of a nation or clan including physical, linguistic and spiritual features. Aboriginal communities' cultural associations with their Country may include or relate to languages, cultural practices, knowledge, songs, stories, art, paths, landforms, flora, fauna and minerals. These cultural associations may include custodial relationships with particular landscapes such as land, sea, sky, rivers as well as the intangible places associated with the Dreaming(s). Custodial relationships are extremely important in determining who may have the capacity to authentically speak for their Country.

Place is a space mapped out by physical or intangible boundaries that individuals or groups of Torres Strait Islander Peoples occupy and regard as their own. It is a space with varying degrees of spirituality.

culture

The customs, habits, beliefs/spirituality, social organisation and ways of life that characterise different groups and communities. Cultural characteristics give a group or individual a sense of who they are and help them make sense of the world in which they live. Culture is a shared system but inherently diverse – there can be individual and group differences within cultures. Everyone has culture – it is a lens through which we see the world.

custodians

In Aboriginal communities, an individual charged with maintaining and passing on particular elements of cultural significance, eg language, stories, songs, rituals and imagery.

See owners

d/Deaf

When referring to deaf people who belong to a linguistic and cultural minority known as the Deaf community, the 'D' may be capitalised in reference to the individual, the group, or the culture in order to accord respect and deference, for example, the Deaf community. When referring simply to audiological status or when cultural affiliation is not known, as in the case of a person with a hearing loss in general, the lowercase 'd', as in 'deaf' is the more common usage.

Deaf

A cultural identity for people with hearing loss who share a common culture and who usually have a shared sign language.

disability

An umbrella term for any or all of the following components:

- impairments: challenges in body function or structure
- activity limitations: difficulties in executing activities

- participation restrictions: challenges an individual may experience in involvement in life situations. (World Health Organization)

diversity

Differences that exist within a group, for example, age, sex, gender, gender expression, sexuality, ethnicity, ability/disability, body shape and composition, culture, religion/spirituality, learning differences, socioeconomic background, values and experiences.

Dreaming

The Dreaming has different meanings for different Aboriginal groups. The Dreaming can be seen as the embodiment of Aboriginal creation which gives meaning to everything; the essence of Aboriginal beliefs about creation and spiritual and physical existence. It establishes the rules governing relationships between the people, the land and all things for Aboriginal Peoples. The Dreaming is linked to the past, the present and the future. Where appropriate, refer to Aboriginal names for the Dreaming.

Elders

The custodians of knowledge and lore. They are chosen and accepted by their own communities as people who have the permission to disclose cultural knowledge and beliefs. Recognised Elders are highly respected people within Aboriginal communities. Proper consultation with local Aboriginal communities will often direct schools to recognised Elders.

first language(s)

The first language(s) that a person learns to speak.

Indigenous

An internationally recognised term for the first peoples of a land. In NSW the term Aboriginal person/Peoples is preferred.

See Aboriginal and Torres Strait Islander Peoples

Indigenous cultural and intellectual property (ICIP)

Includes, but is not limited to, objects, sites, cultural knowledge, cultural expression and the arts, that have been transmitted or continue to be transmitted through generations as belonging to a particular Indigenous group or Indigenous people as a whole or their territory.

See intellectual property

intellectual property

Non-material assets such as forms of cultural expression that belong to a particular individual or community. Intellectual property rights refer to the rights that the law grants to individuals for the protection of creative, intellectual, scientific and industrial activity, such as inventions.

See Indigenous cultural and intellectual property (ICIP), and copyright

Key Word Sign

A communication strategy that incorporates signing with speech. It is used to support language development for people with communication difficulties. Although Key Word Sign uses a simplified form of manual signing, it is different to Auslan, as it is not a signed language.

kinship

A key aspect of Aboriginal cultures and values. It includes the importance of all relationships and of being related to and belonging to the land.

language group

An Aboriginal community identified with a common language, both verbal and nonverbal, and with a particular territory. Used in preference to the term 'tribe'.

language revival

The process and range of strategies for increasing knowledge and use of a language that is no longer spoken fluently across all generations in the context of language loss or language dispossession caused by colonisation. Aboriginal Languages and Torres Strait Islander Languages are being revived through community initiatives, linguistic research and school programs. 'Language revival' may be used as an overarching term that could also include 'reclamation', 'revitalisation', 'renewal' and 'reawakening'.

local Aboriginal community

A local Aboriginal community is constituted by those people who are Aboriginal and who reside in the near locality. Aboriginal communities will have a rich and diverse history that has been seriously affected by dispossession and relations, which sees families with spiritual connection to Country residing beside those who have been forced to move from other locations. The notion of locality is complex and multilayered: schools should seek advice from a range of people and/or organisations representing local interests.

See community(ies)

off-Country

Taking place away from Aboriginal land or Country of origin.

See Country/Place

on-Country

Taking place on Aboriginal land or Country of origin.

See Country/Place

owners

Each Aboriginal Language is recognised as belonging to a particular geographical area and thus to the people who can claim a connection to that area. Aboriginal community members acquire ownership of their language(s) at birth. Language proficiency is not essential for ownership.

See custodians

protocols

The appropriate ways of behaving, communicating and showing respect for diversity of history and culture. This involves appreciation of the knowledge, standing and status of people within the local Aboriginal community and the school community. Protocols inevitably vary between communities, and between people within a community. In establishing a partnership between schools and Aboriginal communities, it is especially important that protocols are acknowledged and respected.

respect

A term used commonly in NSW Aboriginal communities to refer to the way an individual treats others. Showing respect occurs in many ways, such as waiting to speak, listening and demonstrating understanding, not asking too many direct questions, ensuring that people are not made to feel uncomfortable or uneasy, and generally showing regard for others' ideas, beliefs and culture.

sign

Hand signs (or hand talk) used to supplement or replace oral language. Signs form part of nonverbal communication for Aboriginal and Torres Strait Islander Peoples and may be used by people who are hearing, or d/Deaf or hard of hearing. Aboriginal and Torres Strait Islander Sign Languages may be used in some areas. Some Sign Languages may be associated with sacred ceremonial practices.

signage

The use of words, graphic designs and/or symbols used to communicate a message, eg information signs, plaques, warning signs, road signs, signs that show direction.

switch

A type of assistive technology that enables people with cognitive and/or physical disability to access a range of devices, including computers and communication devices. Switches can be activated by touch, or triggered without contact, such as through eye gaze, sound or blowing.

See assistive technology (AT)

yarning circle

Yarning circles are an important cultural practice for Aboriginal and Torres Strait Islander Peoples to learn within the collective group. Knowledge and information are shared in harmony and respect with all individuals.